

An endpoint logarithmic restriction estimate for the integer parabola

Abstract

We prove an endpoint estimate for the discrete Fourier extension operator associated with the integer parabola. For arbitrary complex coefficients,

$$\left\| \sum_{n=1}^N a_n e^{2\pi i(n x + n^2 t)} \right\|_{L^6(\mathbb{T}^2)} \leq C [\log(2N)]^{1/6} \left(\sum_{n=1}^N |a_n|^2 \right)^{1/2},$$

where the torus carries probability Haar measure and C is absolute.

Keywords. Discrete restriction; parabola; sixth moments; non-Archimedean Fourier analysis; logarithmic estimates.

1 Introduction

Let $e(s) = \exp(2\pi i s)$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For $a \in \mathbb{C}^N$, set

$$E_N a(x, t) = \sum_{n=1}^N a_n e(n x + n^2 t), \quad K_{\mathbb{T}}(N) = \sup_{a \neq 0} \frac{\|E_N a\|_{L^6(\mathbb{T}^2)}}{\|a\|_{\ell^2}}. \quad (1.1)$$

All torus integrals are taken with respect to probability Haar measure, and all logarithms are natural. The sixth power of the extension norm is a weighted count of solutions of

$$n_1 + n_2 + n_3 = n_4 + n_5 + n_6, \quad n_1^2 + n_2^2 + n_3^2 = n_4^2 + n_5^2 + n_6^2.$$

This connects the endpoint restriction problem with the arithmetic of the parabola and with periodic Schrödinger estimates.

Bourgain's work on lattice restriction [1] gives the classical lower bound $K_{\mathbb{T}}(N) \gtrsim (\log N)^{1/6}$. Guth, Maldague and Wang [2] obtained a polylogarithmic decoupling estimate for the parabola by a high–low argument. Guo, Li and Yung [3] developed this argument in a non-Archimedean setting and proved

$$K_{\mathbb{T}}(N) \lesssim_{\varepsilon} (\log N)^{2+\varepsilon} \quad (\varepsilon > 0).$$

The exponent $1/6$ is the smallest possible fixed logarithmic exponent. We establish the corresponding upper bound.

Theorem 1.1. *There is an absolute constant C such that, for every integer $N \geq 1$ and every $a \in \mathbb{C}^N$,*

$$\|E_N a\|_{L^6(\mathbb{T}^2)} \leq C [\log(2N)]^{1/6} \|a\|_{\ell^2}. \quad (1.2)$$

Consequently, $K_{\mathbb{T}}(N) \asymp (\log N)^{1/6}$ for $N \geq 2$.

The proof retains the tube geometry and pruning framework used in [3], but estimates the accumulated pruning error through the mass actually removed. An exact fourth-moment identity controls the sum of the high-frequency energies. The high-frequency terms form martingale differences, and a maximal inequality controls all stopping regions at once. The resulting weak estimate is uniform in the depth of the frequency decomposition. A local amplitude decomposition costs one logarithm.

The final summation uses more than this broad estimate alone. Finite character averaging gives an integral recursion in which every child sixth moment has coefficient one. The remaining term has the form $A^4 B^2$, where A and B are the largest two child amplitudes. Within a class with a common terminal mass ratio, scaling down a child of large mass bounds this term by a difference of cubic masses. These differences telescope over the tree. If X_J denotes the finite best constant up to depth J and $D \gtrsim J$, the argument gives $X_J \leq 1 + C(p) \sqrt{DX_J}$, and hence $X_J \lesssim_p D$. The torus embedding belongs to this class for every choice of complex coefficients.

The non-Archimedean argument applies to every fixed odd prime $p > 2$. We use p^2 children at each step and keep the prime dependence of the auxiliary constants explicit. The geometric arguments use $|2|_p = 1$, and hence have the same form at every prime in this range.

Section 2 records the Fourier normalization and local geometry. Section 3 establishes the pruning identities and the total high-frequency energy estimate. The stopping argument and the broad estimates occupy Section 4. Character averaging and parabolic rescaling are given in Section 5, followed by the endpoint closure in Section 6. Section 7 completes the transfer to the torus.

2 Non-Archimedean Fourier analysis and tube geometry

Fix an odd prime $p > 2$. Equip $\mathbb{Q}_p$ with $|p|_p = p^{-1}$ and put $\mathbb{Z}_p = \{x : |x|_p \leq 1\}$. For $z = (x, t) \in \mathbb{Q}_p^2$, write $\|z\|_{p,\max} = \max\{|x|_p, |t|_p\}$. All balls are closed:

$$B(z, r) = \{w \in \mathbb{Q}_p^2 : \|w - z\|_{p,\max} \leq r\}, \quad r \in p^{\mathbb{Z}}.$$

Haar measure is normalized by $\mu(\mathbb{Z}_p^2) = 1$, so $\mu(B(z, r)) = r^2$. Unless another space or measure is indicated, $\|\cdot\|_q$ denotes the $L^q(\mathbb{Q}_p^2, \mu)$ norm, and L^∞ means essential supremum. The notation $\|h\|_{L^\infty(Q)}$ uses the restriction of μ to Q . The torus always carries probability Haar measure.

For nonnegative quantities, $A \lesssim B$ means $A \leq CB$ with an absolute constant, and $A \lesssim_p B$ permits dependence only on p ; $\gtrsim$ denotes the reverse inequality and $\asymp$ the corresponding two-sided comparison. The named constants in the non-Archimedean argument depend only on the fixed prime p , unless stated to be numerical constants. They are independent of the depth, amplitudes and functions. For integrable functions, our convolution convention is

$$(f * g)(z) = \int_{\mathbb{Q}_p^2} f(z - w)g(w) d\mu(w).$$

Since the prime is odd, $|2|_p = 1$. Two intersecting p -adic balls are nested. In particular, distinct balls of radius $r \in p^{\mathbb{Z}}$ have distance strictly greater than r , hence at least pr by the discreteness of the absolute value.

Fix a continuous additive character χ of $\mathbb{Q}_p$ with kernel $\mathbb{Z}_p$. For a Schwartz–Bruhat function, that is, a locally constant compactly supported function, define

$$\widehat{f}(\zeta) = \int_{\mathbb{Q}_p^2} f(z)\chi(-z \cdot \zeta) d\mu(z).$$

We write $\mathcal{S}(\mathbb{Q}_p^2)$ for this function space. For an additive subgroup H , its annihilator is

$$H^\perp = \{\zeta \in \mathbb{Q}_p^2 : \chi(z \cdot \zeta) = 1 \text{ for every } z \in H\}.$$

If $H = A\mathbb{Z}_p^2$ with A invertible, then $H^\perp = A^{-T}\mathbb{Z}_p^2$ and

$$\widehat{\mathbf{1}_{z_0+H}}(\zeta) = \mu(H)\chi(-z_0 \cdot \zeta)\mathbf{1}_{H^\perp}(\zeta), \quad \mu(H)\mu(H^\perp) = 1. \quad (2.1)$$

Indeed, $z \cdot \zeta = y \cdot A^T \zeta$ when $z = Ay$. Since the kernel of χ is $\mathbb{Z}_p$, the character is trivial for every $y \in \mathbb{Z}_p^2$ exactly when $A^T \zeta \in \mathbb{Z}_p^2$. The change of variables gives

$$\mu(H) = |\det A|_p, \quad \mu(H^\perp) = |\det A|_p^{-1}, \quad (H^\perp)^\perp = H.$$

To compute the transform, put $I_H(\zeta) = \int_H \chi(-z \cdot \zeta) d\mu(z)$. Translation invariance gives, for every $h \in H$,

$$I_H(\zeta) = \int_H \chi(-(z+h) \cdot \zeta) d\mu(z) = \chi(-h \cdot \zeta) I_H(\zeta).$$

If $\zeta \notin H^\perp$, one can choose h for which the multiplier differs from one, so $I_H(\zeta) = 0$. If $\zeta \in H^\perp$, the integrand is one and $I_H(\zeta) = \mu(H)$. Translation by z_0 now proves (2.1).

Applying the same integral formula to $H^\perp$ yields inversion for a coset indicator:

$$\begin{aligned} \int \widehat{\mathbf{1}_{z_0+H}}(\zeta) \chi(z \cdot \zeta) d\mu(\zeta) &= \mu(H) \int_{H^\perp} \chi((z-z_0) \cdot \zeta) d\mu(\zeta) \\ &= \mu(H)\mu(H^\perp)\mathbf{1}_H(z-z_0) = \mathbf{1}_{z_0+H}(z). \end{aligned}$$

Every Schwartz–Bruhat function is a finite linear combination of indicators of compact open cosets: local constancy on its compact support supplies a common sufficiently small open subgroup. Its transform has the same property by (2.1). Inversion therefore extends by linearity, and gives

$$\begin{aligned} f(z) &= \int \widehat{f}(\zeta) \chi(z \cdot \zeta) d\mu(\zeta), \\ \int f \overline{g} d\mu &= \int \widehat{f \overline{g}} d\mu, \quad \widehat{f \overline{g}} = \widehat{f} * \widehat{g}. \end{aligned} \quad (2.2)$$

For completeness, substituting inversion for g into the inner product gives

$$\int f(z) \overline{g(z)} d\mu(z) = \int \overline{\widehat{g}(\zeta)} \left(\int f(z) \chi(-z \cdot \zeta) d\mu(z) \right) d\mu(\zeta).$$

The inner integral is $\widehat{f}(\zeta)$, which proves Plancherel. Substituting inversion for g in the transform of $f \overline{g}$ instead gives

$$\begin{aligned} \widehat{f \overline{g}}(\zeta) &= \int \widehat{g}(\nu) \left(\int f(z) \chi(-z \cdot (\zeta - \nu)) d\mu(z) \right) d\mu(\nu) \\ &= \int \widehat{f}(\zeta - \nu) \widehat{g}(\nu) d\mu(\nu). \end{aligned}$$

These changes of order are justified by absolute convergence. In particular, the product formula and $\widehat{\overline{g}(\zeta)} = \overline{\widehat{g}(-\zeta)}$ imply

$$\text{supp } \widehat{f \overline{g}} \subseteq \text{supp } \widehat{f} + \text{supp } \widehat{g}, \quad \text{supp } \widehat{f \overline{g}} \subseteq \text{supp } \widehat{f} - \text{supp } \widehat{g}.$$

Set $R_j = p^{4j}$ for $j \geq 0$, and let $\mathcal{P}_j$ be the partition of $\mathbb{Z}_p$ into cosets of $p^{2j}\mathbb{Z}_p$. Each interval has radius $R_j^{-1/2}$, and each has $b = p^2$ children in $\mathcal{P}_{j+1}$. Define

$$\Xi_j = \{(\xi, \eta) : \xi \in \mathbb{Z}_p, |\eta - \xi^2|_p \leq R_j^{-1}\}, \quad \Theta_j(\tau) = \Xi_j \cap (\tau \times \mathbb{Q}_p).$$

For a frequency interval τ , the projection f_τ is defined by $\widehat{f_\tau} = \mathbf{1}_\tau(\xi) \widehat{f}$. These projections preserve $\mathcal{S}$ and are orthogonal in L^2 for disjoint intervals, since the corresponding Fourier supports have disjoint first-coordinate projections.

Lemma 2.1 (Caps and tubes). *Let $j \in \mathbb{Z}_{\geq 0}$ and $\tau = a + p^{2j}\mathbb{Z}_p \in \mathcal{P}_j$, with $a \in \mathbb{Z}_p$. Then $\Theta_j(\tau) = (a, a^2) + V_j(a)$, where*

$$V_j(a) = \{(u, v) : |u|_p \leq R_j^{-1/2}, |v - 2au|_p \leq R_j^{-1}\},$$

$$W_j(a) := V_j(a)^\perp = \{(x, t) : |x + 2at|_p \leq R_j^{1/2}, |t|_p \leq R_j\}.$$

The tube $W_j(a)$ has measure $R_j^{3/2}$. If $h \in \mathcal{S}(\mathbb{Q}_p^2)$ and $\text{supp } \widehat{h} \subseteq \Theta_j(\tau)$, then $|h|$ is constant on every translate of $W_j(a)$. Restricting h to any such translate preserves its Fourier support in $\Theta_j(\tau)$.

Proof. Write $\xi = a + u$ and $\eta = a^2 + v$. The curvature error becomes

$$\eta - \xi^2 = v - 2au - u^2, \quad |u^2|_p \leq R_j^{-1}.$$

Adding or subtracting a number of norm at most R_j^{-1} preserves the ball of that radius. Thus $|v - 2au - u^2|_p \leq R_j^{-1}$ is equivalent to $|v - 2au|_p \leq R_j^{-1}$, proving the cap identity.

Put $w = v - 2au$. The variables u, w range independently over balls of radii $R_j^{-1/2}, R_j^{-1}$, and

$$xu + tv = (x + 2at)u + tw.$$

The character of this pairing is one for every such u, w exactly when $|x + 2at|_p \leq R_j^{1/2}$ and $|t|_p \leq R_j$. This proves the annihilator formula. The map $(x, t) \mapsto (x + 2at, t)$ has determinant one, so

$$\mu(W_j(a)) = R_j^{1/2}R_j = R_j^{3/2}, \quad \mu(V_j(a)) = R_j^{-3/2}.$$

To prove constancy of the amplitude, let $c = (a, a^2)$ and $w \in W_j(a)$. Inversion, followed by the change of variable $\zeta = c + v$, gives

$$\begin{aligned} h(z + w) &= \int_{V_j(a)} \widehat{h}(c + v) \chi((z + w) \cdot (c + v)) d\mu(v) \\ &= \chi(w \cdot c) \int_{V_j(a)} \widehat{h}(c + v) \chi(z \cdot (c + v)) d\mu(v) = \chi(w \cdot c) h(z). \end{aligned}$$

The second equality uses $\chi(w \cdot v) = 1$. Equivalently, $\chi(-z \cdot c)h(z)$ is invariant under translation by $W_j(a)$, and hence $|h|$ is constant on its cosets.

For a translate $U = z_0 + W_j(a)$, formula (2.1) shows that $\text{supp } \widehat{\mathbf{1}_U} = V_j(a)$. Therefore

$$\text{supp } \widehat{h\mathbf{1}_U} \subseteq (c + V_j(a)) + V_j(a) = c + V_j(a) = \Theta_j(\tau).$$

The cosets of the compact open subgroup $W_j(a)$ partition $\mathbb{Q}_p^2$, and only finitely many meet the compact support of h . Thus the same support conclusion holds when any collection of the occupied tube cosets is retained. $\square$

Lemma 2.2 (Separated bilinear estimate). *Let $\delta = p^{-2s}$, where $s \geq 1$ is an integer. Let $I_1, I_2 \subseteq \mathbb{Z}_p$ be balls of radius at least $\sqrt{\delta}$ and suppose*

$$|\xi_1 - \xi_2|_p \geq \kappa \geq \sqrt{\delta} \quad (\xi_i \in I_i).$$

If $u, v \in \mathcal{S}(\mathbb{Q}_p^2)$ have Fourier supports in $\{\xi \in I_1, |\eta - \xi^2|_p \leq \delta\}$ and $\{\xi \in I_2, |\eta - \xi^2|_p \leq \delta\}$, respectively, then

$$\int |uv|^2 d\mu \leq \frac{\delta^2}{\kappa} \|u\|_2^2 \|v\|_2^2. \quad (2.3)$$

Proof. Partition I_1, I_2 into balls θ, ω of radius $\sqrt{\delta}$, respectively. The corresponding projections satisfy $u = \sum_{\theta} u_{\theta}$ and $v = \sum_{\omega} v_{\omega}$. We first show that the products $u_{\theta}v_{\omega}$ have disjoint Fourier supports as (θ, ω) varies. If two such supports intersect, the convolution support formula supplies points (ξ_i, η_i) , $1 \leq i \leq 4$, with

$$\begin{aligned} \xi_1 + \xi_2 &= \xi_3 + \xi_4, & \eta_1 + \eta_2 &= \eta_3 + \eta_4, \\ \xi_1 \in \theta, & \xi_2 \in \omega, & \xi_3 \in \theta', & \xi_4 \in \omega', & |\eta_i - \xi_i^2|_p &\leq \delta. \end{aligned}$$

Writing $e_i = \eta_i - \xi_i^2$ and using the ultrametric inequality gives

$$|\xi_1^2 + \xi_2^2 - \xi_3^2 - \xi_4^2|_p = |-e_1 - e_2 + e_3 + e_4|_p \leq \delta.$$

The equality of the first-coordinate sums implies the factorization

$$\xi_1^2 + \xi_2^2 - \xi_3^2 - \xi_4^2 = 2(\xi_1 - \xi_3)(\xi_1 - \xi_4).$$

Since $p > 2$, the factor 2 is a unit. The cross separation $|\xi_1 - \xi_4|_p \geq \kappa$ and the factorization give

$$|\xi_1 - \xi_3|_p \leq \frac{\delta}{\kappa} \leq \sqrt{\delta}.$$

The first-coordinate sum also gives

$$|\xi_2 - \xi_4|_p = |\xi_1 - \xi_3|_p \leq \sqrt{\delta}.$$

Two balls of the same radius which contain points this close coincide. Hence $\theta = \theta'$ and $\omega = \omega'$, as required. Plancherel now gives the exact identity

$$\|uv\|_2^2 = \sum_{\theta, \omega} \|u_{\theta}v_{\omega}\|_2^2.$$

Fix one pair, and choose $a_1 \in \theta, a_2 \in \omega$. The cap/tube computation in Lemma 2.1 uses only the bound $|u^2|_p \leq r^2$ and character duality. It therefore applies at every radius $r \in p^{\mathbb{Z}}$, not only at $r = R_j^{-1/2}$. With $r = \sqrt{\delta} = p^{-s}$, including odd s , the relevant tube subgroups are

$$W_i = \{(x, t) : |x + 2a_it|_p \leq \delta^{-1/2}, |t|_p \leq \delta^{-1}\}, \quad \mu(W_i) = \delta^{-3/2}.$$

For translates $U_i = z_i + W_i$, the two transverse constraints require $x + 2a_it$ to belong to a specified ball of radius $\delta^{-1/2}$. The map

$$(x, t) \mapsto (x + 2a_1t, x + 2a_2t)$$

has determinant $2(a_2 - a_1)$. Dropping the constraints on t therefore bounds the intersection by the inverse image of a rectangle of measure δ^{-1} , and gives

$$\mu(U_1 \cap U_2) \leq \frac{\delta^{-1}}{|2(a_1 - a_2)|_p} = \frac{\delta^{-1}}{|a_1 - a_2|_p} \leq \frac{\delta^2}{\kappa} \mu(U_1) \mu(U_2).$$

The squared amplitudes are constant on their respective tube cosets, so they have finite expansions with nonnegative coefficients,

$$|u_{\theta}|^2 = \sum_{U_1} \alpha_{U_1} \mathbf{1}_{U_1}, \quad |v_{\omega}|^2 = \sum_{U_2} \beta_{U_2} \mathbf{1}_{U_2}.$$

Consequently,

$$\begin{aligned} \int |u_{\theta}v_{\omega}|^2 d\mu &= \sum_{U_1, U_2} \alpha_{U_1} \beta_{U_2} \mu(U_1 \cap U_2) \\ &\leq \frac{\delta^2}{\kappa} \left(\sum_{U_1} \alpha_{U_1} \mu(U_1) \right) \left(\sum_{U_2} \beta_{U_2} \mu(U_2) \right) = \frac{\delta^2}{\kappa} \|u_{\theta}\|_2^2 \|v_{\omega}\|_2^2. \end{aligned}$$

Summing over the orthogonal pairs and using orthogonality of the projections in each factor proves

$$\|uv\|_2^2 \leq \frac{\delta^2}{\kappa} \left(\sum_{\theta} \|u_{\theta}\|_2^2 \right) \left(\sum_{\omega} \|v_{\omega}\|_2^2 \right) = \frac{\delta^2}{\kappa} \|u\|_2^2 \|v\|_2^2.$$

□

3 Pruning and square-function energy

Fix $J \geq 1$ and $f \in \mathcal{S}(\mathbb{Q}_p^2)$ with $\text{supp } \widehat{f} \subseteq \Xi_J$. Write

$$S_* = \sum_{\theta \in \mathcal{P}_J} \|f_{\theta}\|_{\infty}^2, \quad M = \|f\|_2^2 = \sum_{\theta \in \mathcal{P}_J} \|f_{\theta}\|_2^2, \quad S = \left\| \sum_{\theta \in \mathcal{P}_J} |f_{\theta}|^2 \right\|_{\infty} \leq S_*. \quad (3.1)$$

The case $S = 0$ means $f = 0$ and will always be understood separately.

Given $\lambda > 0$, start with $F_{J+1, \theta} = f_{\theta}$ and prune downwards. For $j = J, J-1, \dots, 1$, put

$$F_{j, \tau} = F_{j+1, \tau} \mathbf{1}_{\{|F_{j+1, \tau}| \leq \lambda\}} \quad (\tau \in \mathcal{P}_j). \quad (3.2)$$

After defining level j , define every coarser component by summing its level- j descendants:

$$F_{j, \tau} = \sum_{\substack{\sigma \in \mathcal{P}_j \\ \sigma \subseteq \tau}} F_{j, \sigma} \quad (\tau \in \mathcal{P}_{\ell}, \ell < j).$$

At the initial stage use the same convention $F_{J+1, \tau} = f_{\tau}$.

Lemma 3.1 (Support and mass under pruning). *All pruned components are Schwartz–Bruhat functions. For $1 \leq j \leq J$ and $\tau \in \mathcal{P}_j$,*

$$\text{supp } \widehat{F_{j+1, \tau}}, \text{supp } \widehat{F_{j, \tau}} \subseteq \Theta_j(\tau), \quad \|F_{j, \tau}\|_{\infty} \leq \lambda,$$

and

$$\sum_{\tau \in \mathcal{P}_j} \|F_{j, \tau}\|_2^2 \leq \sum_{\tau \in \mathcal{P}_j} \|F_{j+1, \tau}\|_2^2 \leq M.$$

For $\ell < q \leq J$, the coarser component $F_{q, \tau}$, $\tau \in \mathcal{P}_{\ell}$, has Fourier support in $\{\xi \in \tau, |\eta - \xi^2|_p \leq R_q^{-1}\}$.

Proof. At $j = J$, the incoming component is f_{θ} , whose support is in $\Theta_J(\theta)$. Suppose at some level j that the incoming support lies in $\Theta_j(\tau)$. Lemma 2.1 makes its amplitude constant on every level- j tube coset. Thus the cutoff in (3.2) either retains or deletes the entire incoming function on each such coset. Only finitely many occupied cosets occur, and restricting to each preserves the cap. Their finite sum is Schwartz–Bruhat, has support in the same cap, and has amplitude at most λ .

For $j < J$, the incoming component is precisely the sum of the surviving children:

$$F_{j+1, \tau} = \sum_{\substack{\sigma \in \mathcal{P}_{j+1} \\ \sigma \subseteq \tau}} F_{j+1, \sigma}.$$

If the support assertion has been proved at level $j+1$, this sum has Fourier support in

$$\bigcup_{\sigma \subseteq \tau} \Theta_{j+1}(\sigma) = \{(\xi, \eta) : \xi \in \tau, |\eta - \xi^2|_p \leq R_{j+1}^{-1}\} \subseteq \Theta_j(\tau).$$

This establishes the input needed for the next pruning step and completes downward induction. More generally, summing the level- q caps over the descendants of any $\tau \in \mathcal{P}_{\ell}$, $\ell < q$, gives exactly the asserted thickness R_q^{-1} .

At each cutoff, pointwise contraction gives $\|F_{j,\tau}\|_2^2 \leq \|F_{j+1,\tau}\|_2^2$. Regrouping causes no further loss: for $j < J$, disjoint child intervals and Plancherel imply

$$\begin{aligned} \sum_{\tau \in \mathcal{P}_j} \|F_{j+1,\tau}\|_2^2 &= \sum_{\tau \in \mathcal{P}_j} \sum_{\substack{\sigma \in \mathcal{P}_{j+1} \\ \sigma \subseteq \tau}} \|F_{j+1,\sigma}\|_2^2 \\ &= \sum_{\sigma \in \mathcal{P}_{j+1}} \|F_{j+1,\sigma}\|_2^2. \end{aligned}$$

The terminal incoming sum equals M by (3.1). Alternating these exact regrouping identities with the cutoff inequalities proves the mass bounds at every level. $\square$

Define

$$g_j = \sum_{\tau \in \mathcal{P}_j} |F_{j+1,\tau}|^2, \quad \widetilde{g}_j = \sum_{\tau \in \mathcal{P}_j} |F_{j,\tau}|^2.$$

Then $\int g_j \leq M$ and $g_J \leq S$. Let E_j denote averaging on balls of radius $R_j^{1/2}$:

$$E_j h = R_j^{-1} h * \mathbf{1}_{B(0, R_j^{1/2})}.$$

For $j < J$, set $g_j^\ell = E_{j+1} g_j$ and $g_j^h = g_j - g_j^\ell$.

Lemma 3.2 (Low frequencies and difference products). *For $j < J$,*

$$g_j^\ell = \widetilde{g}_{j+1} \leq g_{j+1}, \quad E_j g_j = g_j. \quad (3.3)$$

Moreover, if $u_\sigma = F_{j+1,\sigma}$ for $\sigma \in \mathcal{P}_{j+1}$, then the Fourier supports of $u_\sigma \overline{u_{\sigma'}}$, indexed by distinct ordered pairs $\sigma \neq \sigma'$, are pairwise disjoint.

Proof. The averaging kernel has integral one. Formula (2.1) gives its Fourier transform, and hence

$$\widehat{E_j h}(\zeta) = \mathbf{1}_{B(0, R_j^{-1/2})}(\zeta) \widehat{h}(\zeta).$$

Fix $j < J$, put $r = R_{j+1}^{-1/2}$, and expand one parent into its children:

$$|F_{j+1,\tau}|^2 = \sum_{\sigma \subseteq \tau} |u_\sigma|^2 + \sum_{\substack{\sigma, \sigma' \subseteq \tau \\ \sigma \neq \sigma'}} u_\sigma \overline{u_{\sigma'}}.$$

For a diagonal product, the support is contained in the difference subgroup of the child cap. If $a \in \sigma$, this subgroup satisfies

$$|u|_p \leq r, \quad |v - 2au|_p \leq r^2, \quad |v|_p \leq \max\{|2au|_p, r^2\} \leq r.$$

Here $a \in \mathbb{Z}_p$ and $r \leq 1$. Thus the diagonal support lies in $B(0, r)$. For distinct children, every first frequency coordinate of the difference product has the form $u = \xi - \xi'$ with $\xi \in \sigma$, $\xi' \in \sigma'$. Distinct radius- r balls give $|u|_p > r$, so this support misses $B(0, r)$. The multiplier of E_{j+1} therefore retains exactly the diagonal terms. Summing over parents gives

$$E_{j+1} g_j = \sum_{\sigma \in \mathcal{P}_{j+1}} |F_{j+1,\sigma}|^2 = \widetilde{g}_{j+1} \leq g_{j+1},$$

where the inequality is the pointwise contraction at level $j + 1$.

For the other identity, the incoming parent support lies in $\Theta_j(\tau)$. Its difference subgroup satisfies

$$|u|_p \leq R_j^{-1/2}, \quad |v - 2au|_p \leq R_j^{-1}, \quad |v|_p \leq R_j^{-1/2}.$$

Consequently every $|F_{j+1,\tau}|^2$, and hence g_j , has Fourier support inside the multiplier ball of E_j . This proves $E_j g_j = g_j$.

It remains to distinguish all ordered difference products, including ones whose children belong to different parents. An intersection of the supports associated to (σ_1, σ_2) and (σ_3, σ_4) supplies four frequency points such that

$$\begin{aligned} \xi_1 - \xi_2 &= \xi_3 - \xi_4 = u, & \eta_1 - \eta_2 &= \eta_3 - \eta_4, \\ \xi_i &\in \sigma_i, & \eta_i &= \xi_i^2 + e_i, & |e_i|_p &\leq r^2. \end{aligned}$$

The thickness r^2 is available because these are the pruned level- $(j+1)$ components. Substituting $\xi_2 = \xi_1 - u$ and $\xi_4 = \xi_3 - u$ gives

$$\xi_1^2 - \xi_2^2 - \xi_3^2 + \xi_4^2 = 2u(\xi_1 - \xi_3), \quad |2u(\xi_1 - \xi_3)|_p \leq r^2.$$

Since $\sigma_1 \neq \sigma_2$, the two radius- r balls have separation $|u|_p \geq pr$. Using $|2|_p = 1$ in the preceding inequality gives

$$|\xi_1 - \xi_3|_p \leq \frac{r^2}{|u|_p} \leq \frac{r}{p} < r, \quad |\xi_2 - \xi_4|_p = |\xi_1 - \xi_3|_p < r.$$

The equal-radius ball property forces $\sigma_1 = \sigma_3$ and $\sigma_2 = \sigma_4$. Thus different ordered pairs have disjoint Fourier supports, and their products are orthogonal by Plancherel. $\square$

Lemma 3.3 (Total high-frequency energy). *For every fixed pruning threshold λ ,*

$$\sum_{j=1}^{J-1} \|g_j^h\|_2^2 \leq \frac{b^2}{2} \lambda^2 M. \quad (3.4)$$

Proof. The sum is empty when $J = 1$. For $J \geq 2$, let

$$\begin{aligned} A_j &= \sum_{\tau \in \mathcal{P}_j} \|F_{j+1,\tau}\|_4^4, & B_j &= \sum_{\tau \in \mathcal{P}_j} \|F_{j,\tau}\|_4^4, \\ m_j &= \int g_j, & n_j &= \int \widetilde{g}_j. \end{aligned}$$

Orthogonality in the regrouping identity from Lemma 3.1 gives $m_j = n_{j+1}$ for $j < J$. The cutoff gives $n_j \leq m_j$, while $m_J = M$ and $n_J \leq M$.

Fix $j < J$ and a parent $\tau \in \mathcal{P}_j$, and put $u_\sigma = F_{j+1,\sigma}$ for its children. Write

$$|F_{j+1,\tau}|^2 = L_\tau + H_\tau, \quad L_\tau = \sum_{\sigma \subseteq \tau} |u_\sigma|^2, \quad H_\tau = \sum_{\substack{\sigma, \sigma' \subseteq \tau \\ \sigma \neq \sigma'}} u_\sigma \overline{u_{\sigma'}}.$$

The diagonal support lies inside $B(0, R_{j+1}^{-1/2})$, and the other supports lie outside it. Hence L_τ and H_τ are orthogonal. Define the nonnegative quantity

$$C_\tau = \sum_{\substack{\sigma, \sigma' \subseteq \tau \\ \sigma \neq \sigma'}} \int |u_\sigma|^2 |u_{\sigma'}|^2 d\mu.$$

The ordered-product orthogonality in Lemma 3.2 evaluates the high term, whereas direct expansion of the square evaluates the low term:

$$\|H_\tau\|_2^2 = C_\tau, \quad \|L_\tau\|_2^2 = \sum_{\sigma \subseteq \tau} \|u_\sigma\|_4^4 + C_\tau.$$

Thus the cross products contribute twice to the fourth moment of a parent:

$$\|F_{j+1,\tau}\|_4^4 = \|L_\tau\|_2^2 + \|H_\tau\|_2^2 = \sum_{\sigma \subseteq \tau} \|u_\sigma\|_4^4 + 2C_\tau.$$

Products from different parents correspond to different ordered pairs and are also orthogonal. Since $g_j^h = \sum_\tau H_\tau$, it follows that $\|g_j^h\|_2^2 = \sum_\tau C_\tau$. Summing the last parent identity therefore gives

$$A_j = B_{j+1} + 2\|g_j^h\|_2^2. \quad (3.5)$$

For $j < J$, each incoming parent is the sum of b already pruned children. The triangle inequality and their cutoff bounds give

$$|F_{j+1,\tau}| \leq \sum_{\sigma \subseteq \tau} |F_{j+1,\sigma}| \leq b\lambda, \quad A_1 \leq b^2\lambda^2 m_1.$$

To control the loss at an intermediate pruning step, let $D_{j,\tau} = \{|F_{j+1,\tau}| > \lambda\}$. The retained and removed parts have disjoint physical supports, so

$$\begin{aligned} A_j - B_j &= \sum_{\tau \in \mathcal{P}_j} \int_{D_{j,\tau}} |F_{j+1,\tau}|^4 d\mu, \\ m_j - n_j &= \sum_{\tau \in \mathcal{P}_j} \int_{D_{j,\tau}} |F_{j+1,\tau}|^2 d\mu. \end{aligned}$$

The incoming amplitude bound therefore yields

$$A_j - B_j \leq b^2\lambda^2(m_j - n_j) \quad (2 \leq j < J).$$

Summing (3.5) and grouping each intermediate fourth moment with its pruned counterpart gives

$$\begin{aligned} 2 \sum_{j=1}^{J-1} \|g_j^h\|_2^2 &= \sum_{j=1}^{J-1} (A_j - B_{j+1}) = A_1 - B_J + \sum_{j=2}^{J-1} (A_j - B_j) \\ &\leq b^2\lambda^2 \left(m_1 + \sum_{j=2}^{J-1} (m_j - n_j) \right). \end{aligned}$$

Here $-B_J \leq 0$ was discarded. The mass terms telescope without a factor depending on the number of levels:

$$m_1 + \sum_{j=2}^{J-1} (m_j - n_j) = n_2 + \sum_{j=2}^{J-1} (n_{j+1} - n_j) = n_J \leq M.$$

This proves (3.4). For $J = 2$ the intermediate sums are empty and $m_1 = n_2$ gives the same conclusion. The bound $b\lambda$ was used only for $j < J$, where children have already been pruned; no amplitude bound is imposed on the unpruned terminal functions. $\square$

4 Stopping times and broad estimates

For a nonterminal interval $\tau \in \mathcal{P}_k$, define

$$\begin{aligned} P_\tau(z) &= \max_{\substack{\sigma, \sigma' \in \mathcal{P}_{k+1}, \\ \sigma \neq \sigma', \sigma \subseteq \tau}} |f_\sigma(z)f_{\sigma'}(z)|^{1/2}, \\ U_\tau(z) &= \left(\sum_{\substack{\sigma \in \mathcal{P}_{k+1} \\ \sigma \subseteq \tau}} |f_\sigma(z)|^6 \right)^{1/6}, \quad \mathcal{B}_\tau(T) = \{U_\tau \leq TbP_\tau\}, \quad T \geq 1. \end{aligned} \quad (4.1)$$

These quantities always use the original projections. In this section f , J and p are fixed. We also fix $T \geq 1$; the sets $\mathcal{U}_\alpha$ below depend on T , and the pruned arrays and stopping regions depend on the single threshold $\lambda = \lambda(\alpha, T)$. This dependence is suppressed only within one such estimate. At the root write $P = P_{\mathbb{Z}_p}$ and $U = U_{\mathbb{Z}_p}$. For $\alpha > 0$, let

$$\mathcal{U}_\alpha = \{\alpha \leq P < 2\alpha, U \leq 2Tb\alpha\}.$$

Assuming $S > 0$, choose the pruning threshold

$$\lambda = A_0 T b S / \alpha, \quad A_0 = 1000. \quad (4.2)$$

The quantities defined below refer to this single threshold. In particular, arrays arising from different amplitude levels are never combined.

4.1 The stopping rule and its energy bound

Let

$$\delta_j = g_j - \widetilde{g}_j = \sum_{\sigma \in \mathcal{P}_j} |F_{j+1, \sigma}|^2 \mathbf{1}_{\{|F_{j+1, \sigma}| > \lambda\}}, \quad \mathcal{A}_k = \sum_{j=k}^J \delta_j, \quad \mathcal{A}_{J+1} = 0.$$

For $J \geq 2$, define the disjoint stopping regions in the order $k = J - 1, \dots, 1$ by

$$\Omega_k = \left\{ z \notin \bigcup_{j=k+1}^{J-1} \Omega_j : g_k(z) > 2S \text{ or } \mathcal{A}_k(z) > 4S \right\}, \quad L = \mathbb{Q}_p^2 \setminus \bigcup_{k=1}^{J-1} \Omega_k. \quad (4.3)$$

When $J = 1$, take $L = \mathbb{Q}_p^2$ and an empty family of stopping regions. A point in Ω_k failed both stopping tests at every previously inspected level $j > k$. Since also $g_J \leq S$ and $\mathcal{A}_J = \delta_J \leq g_J \leq S$, on Ω_k we have

$$g_j \leq 2S \quad (k < j \leq J), \quad \mathcal{A}_{k+1} \leq 4S. \quad (4.4)$$

On L , all $g_j \leq 2S$ and $\mathcal{A}_1 \leq 4S$; for $J = 1$ these follow directly from the terminal bounds.

Each Ω_k is a union of balls of radius $R_k^{1/2}$, on which g_k is constant. To verify this, if $a \in \mathbb{Z}_p$ and $w = (x, t) \in B(0, R_j^{1/2})$, then

$$|x + 2at|_p \leq \max(|x|_p, |t|_p) \leq R_j^{1/2}, \quad |t|_p \leq R_j^{1/2} \leq R_j.$$

Thus this ball is contained in $W_j(a)$. Lemmas 2.1 and 3.1 imply that $g_j, \widetilde{g}_j$ and δ_j are constant on its translates. For $j \geq k$, every level- k ball is contained in a level- j ball, so g_k and $\mathcal{A}_k$ are constant on level- k balls. Inductively, every earlier region Ω_j , $j > k$, is a union of these smaller balls as well. Both the exclusion of the earlier regions and the two tests in (4.3) therefore select whole level- k balls.

Lemma 4.1 (Stopping energy). *The stopping regions satisfy*

$$\sum_{k=1}^{J-1} \int_{\Omega_k} g_k^2 d\mu \leq 16 \sum_{j=1}^{J-1} \|g_j^h\|_2^2 \leq 8b^2 \lambda^2 M. \quad (4.5)$$

Proof. There is nothing to prove when $J = 1$. Otherwise choose a compact open ball Q_0 containing the supports of all g_j, g_j^ℓ and g_j^h , with radius at least $R_j^{1/2}$, and put $d\mathbb{P} = d\mu/\mu(Q_0)$ on Q_0 . The finite partitions into balls of radius $R_j^{1/2}$ generate sigma-algebras

$$\mathcal{F}_J \subseteq \mathcal{F}_{J-1} \subseteq \dots \subseteq \mathcal{F}_1.$$

Every averaging ball through a point of Q_0 is contained in Q_0 . Consequently conditional expectation onto $\mathcal{F}_j$ is exactly the averaging operator E_j , restricted to Q_0 . Lemma 3.2 gives

$$g_j^h \text{ is } \mathcal{F}_j\text{-measurable,} \quad E_{j+1}g_j^h = E_{j+1}g_j - E_{j+1}^2g_j = 0.$$

Set $H_J = 0$ and $H_k = \sum_{j=k}^{J-1} g_j^h$. To put these in increasing time order, let

$$\mathcal{G}_i = \mathcal{F}_{J-i}, \quad X_i = H_{J-i} \quad (0 \leq i \leq J-1).$$

The function H_k is $\mathcal{F}_k$ -measurable, while H_{k+1} is $\mathcal{F}_{k+1}$ -measurable. Since $H_k = H_{k+1} + g_k^h$,

$$\mathbb{E}[X_{i+1} \mid \mathcal{G}_i] = E_{J-i}H_{J-i-1} = H_{J-i} = X_i.$$

Thus $(X_i, \mathcal{G}_i)$ is a bounded real martingale. Its differences are orthogonal: for $j < \ell$, the function g_ℓ^h is $\mathcal{F}_{j+1}$ -measurable, and hence

$$\mathbb{E}(g_j^h g_\ell^h) = \mathbb{E}[g_\ell^h \mathbb{E}(g_j^h \mid \mathcal{F}_{j+1})] = 0, \quad \mathbb{E}|H_1|^2 = \sum_{j=1}^{J-1} \mathbb{E}|g_j^h|^2.$$

We recall the maximal inequality for a bounded finite martingale $(X_i)_{i=0}^m$. Set $X^* = \max_i |X_i|$. For $t > 0$, partition $\{X^* > t\}$ into the disjoint first-crossing events

$$C_i(t) = \{|X_i| > t\} \cap \bigcap_{r < i} \{|X_r| \leq t\}, \quad 0 \leq i \leq m.$$

Each $C_i(t)$ belongs to $\mathcal{G}_i$. Conditional Jensen and the martingale property give

$$\begin{aligned} t \mathbb{P}(C_i(t)) &\leq \mathbb{E}[|X_i| \mathbf{1}_{C_i(t)}] \\ &\leq \mathbb{E}[\mathbb{E}(|X_m| \mid \mathcal{G}_i) \mathbf{1}_{C_i(t)}] = \mathbb{E}[|X_m| \mathbf{1}_{C_i(t)}]. \end{aligned}$$

Summing in i and using the layer-cake formula, followed by Tonelli's theorem, yields

$$\begin{aligned} \mathbb{E}[(X^*)^2] &= 2 \int_0^\infty t \mathbb{P}(X^* > t) dt \\ &\leq 2 \int_0^\infty \mathbb{E}[|X_m| \mathbf{1}_{\{X^* > t\}}] dt = 2 \mathbb{E}(|X_m| X^*) \\ &\leq 2 \|X_m\|_2 \|X^*\|_2. \end{aligned}$$

All integrands in the Tonelli step are nonnegative. Dividing by $\|X^*\|_2$ when it is nonzero proves $\|X^*\|_2^2 \leq 4 \|X_m\|_2^2$; when it is zero the result already holds. Apply this to the martingale above, use orthogonality, and multiply by $\mu(Q_0)$. Since the H_k vanish outside Q_0 , we obtain

$$\int \max_k |H_k|^2 d\mu \leq 4 \sum_{j=1}^{J-1} \|g_j^h\|_2^2. \quad (4.6)$$

By (3.3), $g_j^h = g_j - \widetilde{g}_{j+1} = g_j - g_{j+1} + \delta_{j+1}$. Summing this equality gives the exact identity

$$H_k = \sum_{j=k}^{J-1} (g_j - g_{j+1} + \delta_{j+1}) = g_k - g_J + \mathcal{A}_{k+1}. \quad (4.7)$$

If $z \in \Omega_k$ and $g_k(z) > 2S$, then $g_J \leq S$ and $\mathcal{A}_{k+1} \geq 0$ imply $H_k \geq g_k - S > g_k/2$. Otherwise $g_k \leq 2S$ and the stopping test forces $\mathcal{A}_k > 4S$. Since $\delta_k \leq g_k$,

$$\mathcal{A}_{k+1} = \mathcal{A}_k - \delta_k > 4S - g_k \geq 2S, \quad H_k > g_k + S \geq g_k.$$

It follows in both cases that $g_k^2 \leq 4|H_k|^2$ on Ω_k . Disjointness of the stopping regions now gives

$$\begin{aligned} \sum_k \int_{\Omega_k} g_k^2 &\leq 4 \sum_k \int \mathbf{1}_{\Omega_k} |H_k|^2 \leq 4 \int \max_k |H_k|^2 \\ &\leq 16 \sum_{j=1}^{J-1} \|g_j^h\|_2^2 \leq 8b^2 \lambda^2 M, \end{aligned}$$

where the last step is Lemma 3.3. □

Lemma 4.2 (Pruning error). *For every $\tau \in \mathcal{P}_1$,*

$$|f_\tau - F_{k+1,\tau}| \leq \frac{4S}{\lambda} \quad \text{on } \Omega_k, \quad |f_\tau - F_{1,\tau}| \leq \frac{4S}{\lambda} \quad \text{on } L.$$

On $\mathcal{U}_\alpha \cap \Omega_k$, some distinct $\tau, \tau' \in \mathcal{P}_1$ satisfy

$$|F_{k+1,\tau} F_{k+1,\tau'}| \geq \frac{3}{4} \alpha^2.$$

The analogous assertion holds on $\mathcal{U}_\alpha \cap L$ with $F_{1,\tau}$.

Proof. For $j \geq 1$, denote the level- j descendants of τ by $\mathcal{D}_j(\tau) = \{\sigma \in \mathcal{P}_j : \sigma \subseteq \tau\}$, so $\mathcal{D}_1(\tau) = \{\tau\}$. The grouping convention for pruned components gives

$$F_{j+1,\tau} - F_{j,\tau} = \sum_{\sigma \in \mathcal{D}_j(\tau)} (F_{j+1,\sigma} - F_{j,\sigma}).$$

Because $F_{J+1,\tau} = f_\tau$, telescoping over the pruning stages yields

$$f_\tau - F_{k+1,\tau} = \sum_{j=k+1}^J \sum_{\sigma \in \mathcal{D}_j(\tau)} (F_{j+1,\sigma} - F_{j,\sigma}).$$

At an individual stage the deleted term is exactly

$$F_{j+1,\sigma} - F_{j,\sigma} = F_{j+1,\sigma} \mathbf{1}_{\{|F_{j+1,\sigma}| > \lambda\}}.$$

On its support, $|F_{j+1,\sigma}| \leq \lambda^{-1} |F_{j+1,\sigma}|^2$. The triangle inequality, followed by enlarging the descendant sum to all of $\mathcal{P}_j$, therefore gives

$$\begin{aligned} |f_\tau - F_{k+1,\tau}| &\leq \lambda^{-1} \sum_{j=k+1}^J \sum_{\sigma \in \mathcal{D}_j(\tau)} |F_{j+1,\sigma}|^2 \mathbf{1}_{\{|F_{j+1,\sigma}| > \lambda\}} \\ &\leq \lambda^{-1} \sum_{j=k+1}^J \delta_j = \lambda^{-1} \mathcal{A}_{k+1}. \end{aligned}$$

Starting the same finite telescope at $j = 1$ gives $|f_\tau - F_{1,\tau}| \leq \mathcal{A}_1/\lambda$. The budgets on Ω_k and L prove both error bounds, including the case $J = 1$ on L .

Fix $z \in \mathcal{U}_\alpha$ in one of these regions and choose a pair attaining the finite maximum defining $P(z)$. Then

$$|f_\tau f_{\tau'}| = P^2 \geq \alpha^2, \quad |f_\tau|, |f_{\tau'}| \leq U \leq 2Tb\alpha.$$

Write G_τ for the relevant approximation, either $F_{k+1,\tau}$ or $F_{1,\tau}$, and put $e_\tau = f_\tau - G_\tau$. Both errors have modulus at most $E_\alpha = 4S/\lambda = 4\alpha/(A_0 T b)$. Expanding the product before taking absolute values gives

$$G_\tau G_{\tau'} - f_\tau f_{\tau'} = -e_\tau f_{\tau'} - e_{\tau'} f_\tau + e_\tau e_{\tau'},$$

and hence

$$\begin{aligned} |G_\tau G_{\tau'} - f_\tau f_{\tau'}| &\leq 4Tb\alpha E_\alpha + E_\alpha^2 \\ &\leq \left(\frac{16}{A_0} + \frac{16}{A_0^2 T^2 b^2} \right) \alpha^2 < \frac{1}{4} \alpha^2. \end{aligned}$$

Here $A_0 = 1000$, $T \geq 1$ and $b = p^2 \geq 4$. The reverse triangle inequality now gives $|G_\tau G_{\tau'}| \geq 3\alpha^2/4$, as required. $\square$

4.2 Weak and strong broad estimates

Lemma 4.3 (Localization on stopping balls). *Let $1 \leq k < J$ and let Q be a ball of radius $R_k^{1/2}$. For distinct $\tau, \tau' \in \mathcal{P}_1$,*

$$\int_Q |F_{k+1,\tau} F_{k+1,\tau'}|^2 \leq \frac{b}{\mu(Q)} \left(\int_Q |F_{k+1,\tau}|^2 \right) \left(\int_Q |F_{k+1,\tau'}|^2 \right). \quad (4.8)$$

Moreover,

$$\sum_{\tau \in \mathcal{P}_1} \int_Q |F_{k+1,\tau}|^2 = \sum_{\sigma \in \mathcal{P}_k} \int_Q |F_{k+1,\sigma}|^2 = \int_Q g_k. \quad (4.9)$$

Proof. Put $r = R_k^{-1/2}$. By (2.1), $\widehat{\mathbf{1}}_Q$ is supported in $B(0, r)$, and $\mu(Q) = R_k$. Suppose first that $k \geq 2$. The incoming root components have curvature thickness R_{k+1}^{-1} by Lemma 3.1. If (ξ, η) is in their Fourier support and $(u, v) \in B(0, r)$, then

$$\begin{aligned} \eta + v - (\xi + u)^2 &= (\eta - \xi^2) + v - 2\xi u - u^2, \\ |\eta + v - (\xi + u)^2|_p &\leq \max(R_{k+1}^{-1}, |v|_p, |2\xi u|_p, |u|_p^2) \leq r. \end{aligned}$$

In the last inequality, $\xi \in \mathbb{Z}_p$, $r \leq 1$, and $R_{k+1}^{-1} \leq r$ were used. Also $|u|_p \leq r \leq p^{-2}$, so the first coordinate remains in its original level-1 interval. Thus $F_{k+1,\tau} \mathbf{1}_Q$ and $F_{k+1,\tau'} \mathbf{1}_Q$ satisfy Lemma 2.2 with

$$\delta = r = p^{-2k}, \quad \sqrt{\delta} = p^{-k} \leq p^{-2}, \quad \kappa = p^{-1}.$$

Indeed, distinct level-1 intervals have radius p^{-2} and separation at least p^{-1} . Applying that lemma gives (4.8), because

$$\frac{\delta^2}{\kappa} = pR_k^{-1} \leq p^2 R_k^{-1} = \frac{b}{\mu(Q)}, \quad \|(F_{k+1,\tau} \mathbf{1}_Q)(F_{k+1,\tau'} \mathbf{1}_Q)\|_2^2 = \int_Q |F_{k+1,\tau} F_{k+1,\tau'}|^2.$$

When $k = 1$, both incoming amplitudes are constant on Q by Lemma 2.1. If their values are a and a' , the two sides with coefficient $1/\mu(Q)$ are both $a^2(a')^2\mu(Q)$; increasing the coefficient to $b/\mu(Q)$ proves the assertion.

For the mass identity, a level- k interval is a coset of the additive ball of radius r . Therefore

$$\mathbf{1}_\sigma(\xi + u) = \mathbf{1}_\sigma(\xi) \quad (\sigma \in \mathcal{P}_k, |u|_p \leq r).$$

Inside convolution with $\widehat{\mathbf{1}}_Q$, the multiplier $\mathbf{1}_\sigma$ can consequently be moved to the unlocalized factor. In particular, for $\sigma \subseteq \tau$,

$$(F_{k+1,\tau} \widehat{\mathbf{1}}_Q)_\sigma = (\mathbf{1}_\sigma \widehat{F_{k+1,\tau}}) * \widehat{\mathbf{1}}_Q = F_{k+1,\sigma} \widehat{\mathbf{1}}_Q.$$

The localized level- k descendants have disjoint first-coordinate supports. Plancherel and the descendant decomposition now imply

$$\begin{aligned} \sum_{\tau \in \mathcal{P}_1} \int_Q |F_{k+1,\tau}|^2 &= \sum_{\tau \in \mathcal{P}_1} \left\| \sum_{\substack{\sigma \in \mathcal{P}_k \\ \sigma \subseteq \tau}} F_{k+1,\sigma} \mathbf{1}_Q \right\|_2^2 \\ &= \sum_{\tau \in \mathcal{P}_1} \sum_{\substack{\sigma \in \mathcal{P}_k \\ \sigma \subseteq \tau}} \|F_{k+1,\sigma} \mathbf{1}_Q\|_2^2 = \int_Q g_k. \end{aligned}$$

Each level- k interval has exactly one level-1 parent, which accounts for the last equality and proves (4.9). $\square$

Proposition 4.4 (Weak broad estimate). *For every $T \geq 1$ and $\alpha > 0$,*

$$\alpha^6 \mu(\mathcal{U}_\alpha) \leq C_W T^2 S_*^2 M, \quad C_W = \frac{128}{9} A_0^2 b^5 + 32. \quad (4.10)$$

Proof. For $f = 0$ the set $\mathcal{U}_\alpha$ is empty. If $J = 1$, the root children are the terminal intervals, so $P^2 \leq g_1 \leq S \leq S_*$. Consequently

$$\alpha^6 \mu(\mathcal{U}_\alpha) \leq \int_{\mathcal{U}_\alpha} P^6 \leq \int g_1^3 \leq S_*^2 \int g_1 = S_*^2 M.$$

Assume $J \geq 2$ and use the threshold (4.2). Lemma 4.2 implies

$$\frac{9}{16} \alpha^4 \mathbf{1}_{\mathcal{U}_\alpha \cap \Omega_k} \leq \mathbf{1}_{\Omega_k} \sum_{\tau \neq \tau' \in \mathcal{P}_1} |F_{k+1,\tau} F_{k+1,\tau'}|^2.$$

Let Q_k be the partition of Ω_k into balls of radius $R_k^{1/2}$, and set $m_\tau(Q) = \int_Q |F_{k+1,\tau}|^2$. Integrating the preceding pointwise inequality and applying (4.8) gives

$$\begin{aligned} \frac{9}{16} \alpha^4 \mu(\mathcal{U}_\alpha \cap \Omega_k) &\leq \sum_{Q \in Q_k} \sum_{\tau \neq \tau'} \int_Q |F_{k+1,\tau} F_{k+1,\tau'}|^2 \\ &\leq b \sum_{Q \in Q_k} \frac{1}{\mu(Q)} \sum_{\tau \neq \tau'} m_\tau(Q) m_{\tau'}(Q). \end{aligned}$$

All local masses are nonnegative, so the sum over distinct pairs is bounded by $(\sum_\tau m_\tau(Q))^2$. The local-mass identity and the constancy of g_k on Q then yield

$$\begin{aligned} b \sum_{Q \in Q_k} \frac{1}{\mu(Q)} \sum_{\tau \neq \tau'} m_\tau(Q) m_{\tau'}(Q) &\leq b \sum_{Q \in Q_k} \frac{1}{\mu(Q)} \left(\int_Q g_k \right)^2 \\ &= b \sum_{Q \in Q_k} \int_Q g_k^2 = b \int_{\Omega_k} g_k^2. \end{aligned}$$

For the middle equality, if $g_k = c_Q$ on Q , then $\mu(Q)^{-1} (c_Q \mu(Q))^2 = c_Q^2 \mu(Q)$. Summing in k and applying (4.5), we conclude that

$$\begin{aligned} \alpha^6 \mu \left(\mathcal{U}_\alpha \cap \bigcup_k \Omega_k \right) &\leq \frac{16}{9} b \alpha^2 \sum_k \int_{\Omega_k} g_k^2 \\ &\leq \frac{128}{9} b^3 \alpha^2 \lambda^2 M = \frac{128}{9} A_0^2 b^5 T^2 S_*^2 M. \end{aligned}$$

Disjointness was used to sum the measures. Compact support ensures that the displayed ball sums have only finitely many nonzero terms.

On $\mathcal{U}_\alpha \cap L$, the surviving pair from Lemma 4.2 obeys

$$\frac{3}{4}\alpha^2 \leq |F_{1,\tau}F_{1,\tau'}| \leq |F_{1,\tau}|^2 + |F_{1,\tau'}|^2 \leq \widetilde{g}_1 \leq g_1 \leq 2S.$$

Cubing the inequality $\alpha^2 \leq 4g_1/3$ and then using $g_1^2 \leq 4S^2$ gives

$$\alpha^6 \leq \frac{64}{27}g_1^3 \leq \frac{256}{27}S^2g_1, \quad \alpha^6\mu(\mathcal{U}_\alpha \cap L) \leq \frac{256}{27}S^2M \leq 32S^2M.$$

The bound $\int g_1 \leq M$ follows from pruning. Adding the two disjoint parts and using $S \leq S_*$ and $T \geq 1$ gives exactly (4.10) with the stated C_W . $\square$

Proposition 4.5 (Strong broad estimate). *Let $D \geq \max\{2, 4J \log p\}$. Uniformly for $T \geq 1$,*

$$\int_{\mathcal{B}_{\mathbb{Z}_p}(T)} P^6 d\mu \leq C_B D T^2 S_*^2 M, \quad C_B = 1 + 256C_W. \quad (4.11)$$

Proof. Partition $\mathbb{Q}_p^2$ into balls Q of radius R_J and set $h = f\mathbf{1}_Q$. The Fourier transform of the cutoff is supported in $B(0, R_J^{-1})$. For $(\xi, \eta) \in \Xi_J$ and $|u|_p, |v|_p \leq R_J^{-1}$,

$$|\eta + v - (\xi + u)^2|_p \leq \max(R_J^{-1}, |v|_p, |2\xi u|_p, |u|_p^2) \leq R_J^{-1}.$$

Thus $\text{supp } \widehat{h} \subseteq \Xi_J$. Moreover $|u|_p \leq R_J^{-1} \leq R_j^{-1/2}$ for $0 \leq j \leq J$, so addition of u preserves every level- j interval. Moving its indicator through the Fourier convolution, as in the preceding localization lemma, gives

$$h_\tau = f_\tau \mathbf{1}_Q \quad (\tau \in \mathcal{P}_j, 0 \leq j \leq J). \quad (4.12)$$

In particular, $P(h) = P$ and $U(h) = U$ on Q . Write

$$M_Q = \|h\|_2^2, \quad A_Q = \max_{\theta \in \mathcal{P}_J} \|f_\theta\|_{L^\infty(Q)}, \quad S_Q = \sum_{\theta \in \mathcal{P}_J} \|f_\theta\|_{L^\infty(Q)}^2.$$

Then $S_*(h) = S_Q$, and orthogonality of the projections in (4.12) gives

$$M_Q = \sum_{\theta \in \mathcal{P}_J} \|h_\theta\|_2^2 = \sum_{\theta \in \mathcal{P}_J} \int_Q |f_\theta|^2.$$

If $A_Q = 0$, every h_θ vanishes and this ball contributes zero. Otherwise choose $\theta_0 = a + p^{2J}\mathbb{Z}_p$ and $z_* \in Q$ at which $|f_{\theta_0}(z_*)| = A_Q$. Such a maximum is attained because the functions are locally constant and Q is compact. For $w = (x, t) \in W_J(a)$,

$$|t|_p \leq R_J, \quad |x|_p \leq \max(|x + 2at|_p, |2at|_p) \leq \max(R_J^{1/2}, R_J) = R_J.$$

Hence $z_* + W_J(a) \subseteq Q$. Lemma 2.1 makes the amplitude A_Q constant on this entire tube, which has measure $R_J^{3/2}$. Consequently,

$$M_Q \geq A_Q^2 R_J^{3/2}. \quad (4.13)$$

Indeed, this lower bound is obtained by retaining only θ_0 in the preceding orthogonal mass sum and integrating over $z_* + W_J(a)$.

Each root child has $b^{J-1} \leq b^J = R_J^{1/2}$ terminal descendants. The triangle inequality applied to their sum gives $|h_\tau| \leq R_J^{1/2} A_Q$, and therefore $P(h) \leq R_J^{1/2} A_Q$. At the other end of the amplitude range, $\mu(Q) = R_J^2$, $S_Q \geq A_Q^2$ and (4.13) show that

$$\begin{aligned} \int_Q P(h)^6 \mathbf{1}_{\{P(h) < R_J^{-1/2} A_Q\}} &\leq R_J^2 (R_J^{-1/2} A_Q)^6 = R_J^{-1} A_Q^6 \\ &\leq R_J^{-5/2} S_Q^2 M_Q \leq S_Q^2 M_Q. \end{aligned}$$

To estimate the remaining amplitudes, put

$$\alpha_\ell = 2^\ell R_J^{-1/2} A_Q, \quad 0 \leq \ell \leq \lceil \log_2 R_J \rceil.$$

These half-open intervals $[\alpha_\ell, 2\alpha_\ell)$ cover the full range from $R_J^{-1/2} A_Q$ through $R_J^{1/2} A_Q$, including the possible upper endpoint. Their number satisfies

$$\lceil \log_2 R_J \rceil + 1 \leq \frac{\log R_J}{\log 2} + 2 \leq \frac{D}{\log 2} + 2 \leq 4D,$$

using $\log R_J = 4J \log p \leq D$ and $D \geq 2$. For the portion of the ℓ th interval in $Q \cap \mathcal{B}_{\mathbb{Z}_p}(T)$, broadness implies

$$U(h) \leq T b P(h) < 2T b \alpha_\ell.$$

It is thus contained in $\mathcal{U}_{\alpha_\ell}(h)$. Proposition 4.4, applied separately to h at this amplitude, gives

$$\begin{aligned} \int_{Q \cap \mathcal{B}_{\mathbb{Z}_p}(T)} P^6 \mathbf{1}_{\{\alpha_\ell \leq P < 2\alpha_\ell\}} &\leq (2\alpha_\ell)^6 \mu(\mathcal{U}_{\alpha_\ell}(h)) \\ &\leq 64 C_W T^2 S_Q^2 M_Q. \end{aligned}$$

Adding the small-amplitude contribution and the at most $4D$ intervals proves

$$\int_{Q \cap \mathcal{B}_{\mathbb{Z}_p}(T)} P^6 \leq (1 + 256 C_W D T^2) S_Q^2 M_Q \leq (1 + 256 C_W) D T^2 S_Q^2 M_Q.$$

Finally, $S_Q \leq S_*$, and disjointness of the spatial balls gives $\sum_Q M_Q = \sum_Q \int_Q |f|^2 = M$. Summing the last inequality yields (4.11). Only finitely many balls contribute, and the uniformity in α of the weak estimate permits each ball and each interval to use its own pruning threshold. $\square$

5 Character averaging and parabolic rescaling

The next recursion is an integral inequality. Its coefficient on each child sixth moment is exactly one.

Lemma 5.1 (Character averaging). *Let $\omega = \chi(1/b)$ and $C_0 = b^6$. For $z_0, \dots, z_{b-1} \in \mathbb{C}$, let $A \geq B$ be their largest and second largest moduli. Then*

$$\frac{1}{b} \sum_{r=0}^{b-1} \left| \sum_{i=0}^{b-1} \omega^{ri} z_i \right|^6 \leq \sum_{i=0}^{b-1} |z_i|^6 + C_0 A^4 B^2. \quad (5.1)$$

Consequently, at every nonterminal node τ ,

$$\int |f_\tau|^6 d\mu \leq \sum_{\sigma \text{ child of } \tau} \int |f_\sigma|^6 d\mu + C_0 \int Q_\tau d\mu, \quad (5.2)$$

where

$$Q_\tau = \max_{\substack{\sigma \neq \sigma' \\ \sigma, \sigma' \text{ children of } \tau}} |f_\sigma|^4 |f_{\sigma'}|^2 = A_\tau^4 B_\tau^2 = A_\tau^2 P_\tau^4$$

and $A_\tau \geq B_\tau$ are the two largest child moduli.

Proof. Since χ has kernel $\mathbb{Z}_p$, we have $\omega^b = \chi(1) = 1$ and $\omega^p = \chi(1/p) \neq 1$. Thus ω has order $b = p^2$. For every integer d , the finite geometric sum gives

$$\frac{1}{b} \sum_{r=0}^{b-1} \omega^{rd} = \begin{cases} 1, & d \equiv 0 \pmod{b}, \\ 0, & d \not\equiv 0 \pmod{b}. \end{cases}$$

Write the sixth power as the product of three copies of the sum and three copies of its conjugate. Averaging each monomial gives the exact identity

$$\begin{aligned} \frac{1}{b} \sum_{r=0}^{b-1} \left| \sum_{i=0}^{b-1} \omega^{ri} z_i \right|^6 \\ = \sum_{i_1, \dots, i_6=0}^{b-1} z_{i_1} z_{i_2} z_{i_3} \overline{z_{i_4} z_{i_5} z_{i_6}} \mathbf{1}_{\{i_1+i_2+i_3-i_4-i_5-i_6 \equiv 0 \pmod{b}\}}. \end{aligned}$$

There are three possible types of monomial to consider. If all six indices equal i , the term survives and equals $|z_i|^6$, with coefficient one. If five indices equal i and the remaining one equals $j \neq i$, the signed index sum is $j - i$ or $i - j$. This is nonzero modulo b because $i, j \in \{0, \dots, b-1\}$, so the term vanishes in the average. Every other surviving monomial contains each label at most four times.

Choose one index i_* with $|z_{i_*}| = A$. All other moduli are at most B , including when the maximum is attained more than once. If $A > 0$ and $B > 0$, a remaining monomial in which i_* occurs m times, $0 \leq m \leq 4$, has modulus at most

$$A^m B^{6-m} = A^4 B^2 (B/A)^{4-m} \leq A^4 B^2.$$

If $A = 0$, all terms vanish. If $B = 0$, only one coefficient can be nonzero, and every remaining monomial vanishes. Hence the same bound holds in these cases. The remainder is real, since it is the difference between the averaged sixth power and its diagonal part. Bounding it above by the sum of the moduli of its at most b^6 monomials proves (5.1).

To apply this scalar identity, write $\tau = a + \rho \mathbb{Z}_p$, where $\rho = p^{2k}$, and label its children by $\sigma_i = a + \rho i + b \rho \mathbb{Z}_p$. If $\xi \in \sigma_i$, then $\xi = a + \rho i + b \rho u$ for some $u \in \mathbb{Z}_p$. Consequently, for $r \in \{0, \dots, b-1\}$,

$$\chi(r\xi/(b\rho)) = \chi(ar/(b\rho))\chi(ri/b)\chi(ru) = \chi(ar/(b\rho))\omega^{ri}.$$

Fourier inversion and the first-coordinate support of f_{σ_i} now yield

$$\begin{aligned} f_{\sigma_i}(x + r/(b\rho), t) &= \int \widehat{f_{\sigma_i}}(\xi, \eta) \chi(x\xi + t\eta) \chi(r\xi/(b\rho)) d\mu(\xi, \eta) \\ &= \chi(ar/(b\rho)) \omega^{ri} f_{\sigma_i}(x, t). \end{aligned}$$

The prefactor is common to all children and has modulus one. Summing the children, using translation invariance for each r , and then averaging gives

$$\int |f_\tau|^6 d\mu = \int \frac{1}{b} \sum_{r=0}^{b-1} \left| \sum_{i=0}^{b-1} \omega^{ri} f_{\sigma_i}(x, t) \right|^6 d\mu(x, t).$$

Apply (5.1) pointwise to obtain (5.2). For completeness, among ordered products of two distinct child moduli, the largest is $A_\tau^4 B_\tau^2$: putting the largest modulus in the squared position instead gives $B_\tau^4 A_\tau^2 \leq A_\tau^4 B_\tau^2$, and a pair omitting it gives at most B_τ^6 . Also $P_\tau^2 = A_\tau B_\tau$, proving the stated formula for Q_τ . All sums and averages are finite, and all integrals are finite because the components are Schwartz–Bruhat functions. $\square$

For a node $\tau \in \mathcal{P}_k$, set

$$S_\tau = \sum_{\substack{\theta \in \mathcal{P}_J \\ \theta \subseteq \tau}} \|f_\theta\|_\infty^2, \quad M_\tau = \sum_{\substack{\theta \in \mathcal{P}_J \\ \theta \subseteq \tau}} \|f_\theta\|_2^2 = \|f_\tau\|_2^2.$$

Both quantities are additive over the children, since the terminal descendants partition the parent and have disjoint first-coordinate Fourier supports.

Lemma 5.2 (Rescaling a node). *Let $D \geq \max\{2, 4J \log p\}$. For every $T \geq 1$, $k < J$, and $\tau \in \mathcal{P}_k$,*

$$\int_{\mathcal{B}_\tau(T)} P_\tau^6 d\mu \leq C_B D T^2 S_\tau^2 M_\tau. \quad (5.3)$$

The same D , T and C_B may be used at every node.

Proof. Write $\tau = a + \rho\mathbb{Z}_p$ with $\rho = p^{2k}$ and define

$$h(x, t) = \chi(-a\rho^{-1}x + a^2\rho^{-2}t) f_\tau(\rho^{-1}x - 2a\rho^{-2}t, \rho^{-2}t). \quad (5.4)$$

This change of variables preserves the Schwartz–Bruhat class. To compute its Fourier transform, write $h(z) = \chi(\beta \cdot z) f_\tau(Bz)$, where

$$B = \begin{pmatrix} \rho^{-1} & -2a\rho^{-2} \\ 0 & \rho^{-2} \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} \rho & 2a\rho \\ 0 & \rho^2 \end{pmatrix}, \quad \beta = \begin{pmatrix} -a\rho^{-1} \\ a^2\rho^{-2} \end{pmatrix}.$$

Changing variables from z to $w = Bz$ in the Fourier integral gives

$$\begin{aligned} \widehat{h}(\zeta) &= |\det B|_p^{-1} \int f_\tau(w) \chi(-w \cdot B^{-T}(\zeta - \beta)) d\mu(w) \\ &= |\rho|_p^3 \widehat{f_\tau}(B^{-T}(\zeta - \beta)). \end{aligned}$$

The inverse transpose and the modulation together give the affine frequency map

$$B^{-T} = \begin{pmatrix} \rho & 0 \\ 2a\rho & \rho^2 \end{pmatrix}, \quad B^{-T} \left(\begin{pmatrix} \xi \\ \eta \end{pmatrix} - \beta \right) = \begin{pmatrix} a + \rho\xi \\ a^2 + 2a\rho\xi + \rho^2\eta \end{pmatrix}.$$

In particular,

$$\widehat{h}(\xi, \eta) = |\rho|_p^3 \widehat{f_\tau}(a + \rho\xi, a^2 + 2a\rho\xi + \rho^2\eta).$$

Denote the two arguments of $\widehat{f_\tau}$ by ξ_0, η_0 . On its support, $\xi_0 \in a + \rho\mathbb{Z}_p$, so $\xi = (\xi_0 - a)/\rho \in \mathbb{Z}_p$. Moreover,

$$\eta_0 - \xi_0^2 = a^2 + 2a\rho\xi + \rho^2\eta - (a + \rho\xi)^2 = \rho^2(\eta - \xi^2).$$

Since $|\rho|_p = p^{-2k}$ and $|\eta_0 - \xi_0^2|_p \leq R_J^{-1}$, this implies

$$|\eta - \xi^2|_p \leq |\rho|_p^{-2} R_J^{-1} = p^{4k} p^{-4J} = R_{J-k}^{-1}.$$

Thus $\text{supp } \widehat{h} \subseteq \Xi_{J-k}$. If $\sigma \subseteq \tau$ is a level- $(k+j)$ interval, its image under $\xi_0 \mapsto (\xi_0 - a)/\rho$ is a coset of $p^{2j}\mathbb{Z}_p$. Applying the Fourier formula with the corresponding first-coordinate cutoff shows that all descendant components undergo the same modulation and matrix B .

The determinant satisfies $|\det B|_p = |\rho|_p^{-3} = R_k^{3/2}$. Since B is bijective and the modulation has modulus one, for $1 \leq q < \infty$ we obtain

$$\|h\|_q^q = R_k^{-3/2} \|f_\tau\|_q^q, \quad S_*(h) = S_\tau, \quad \|h\|_2^2 = R_k^{-3/2} M_\tau. \quad (5.5)$$

The L^∞ norm of each corresponding component is unchanged. In particular, the children of τ become the root children of h , so

$$P(h)(z) = P_\tau(Bz), \quad U(h)(z) = U_\tau(Bz), \quad \mathcal{B}_{\mathbb{Z}_p}^h(T) = B^{-1}\mathcal{B}_\tau(T),$$

where the superscript indicates that the root broad set is formed from h . Changing variables gives

$$\int_{\mathcal{B}_{\mathbb{Z}_p}^h(T)} P(h)^6 d\mu = R_k^{-3/2} \int_{\mathcal{B}_\tau(T)} P_\tau^6 d\mu.$$

The remaining depth is $J - k \geq 1$, and $D \geq \max\{2, 4(J - k) \log p\}$. Proposition 4.5, applied to h with the same T and D , bounds the left side by $C_B D T^2 S_\tau^2 R_k^{-3/2} M_\tau$. Cancelling $R_k^{-3/2} > 0$ proves (5.3). The calculation includes $k = 0$. $\square$

Proposition 5.3 (A general mixed estimate). *For every $f \in \mathcal{S}(\mathbb{Q}_p^2)$ with $\text{supp } \widehat{f} \subseteq \Xi_J$ and $D \geq \max\{2, 4J \log p\}$,*

$$\int |f|^6 d\mu \leq C D^3 S_*^2 M, \quad (5.6)$$

where C depends only on p .

Proof. Choose $T = (C_0 D)^{1/4} \geq 1$. At a node, write $A = A_\tau$, $P = P_\tau$ and $U = U_\tau$. The definitions give $A \leq U$ and $U^6 = \sum_{\sigma \text{ child}} |f_\sigma|^6$. Outside $\mathcal{B}_\tau(T)$ we also have $P < U/(Tb)$, so

$$C_0 Q_\tau = C_0 A^2 P^4 \leq C_0 (Tb)^{-4} U^6 = b^{-4} D^{-1} U^6 \leq D^{-1} U^6.$$

On the broad set we use Young's inequality with its explicit constant. For $x, y \geq 0$ and $\varepsilon > 0$, we claim

$$C_0 x^{1/3} y^{2/3} \leq \varepsilon x + C_Y \varepsilon^{-1/2} y, \quad C_Y = \frac{2C_0^{3/2}}{3\sqrt{3}}.$$

If $y = 0$, the inequality holds directly. If $y > 0$, put $r = (x/y)^{1/3}$. After division by y , it suffices to maximize $C_0 r - \varepsilon r^3$ for $r \geq 0$. Its derivative is $C_0 - 3\varepsilon r^2$, so the maximum occurs at $r_0 = (C_0/(3\varepsilon))^{1/2}$ and equals

$$C_0 r_0 - \varepsilon r_0^3 = \frac{2}{3} C_0 r_0 = \frac{2C_0^{3/2}}{3\sqrt{3}} \varepsilon^{-1/2}.$$

Apply this inequality with $x = A^6$, $y = P^6$, and $\varepsilon = D^{-1}$. Since $Q_\tau = (A^6)^{1/3} (P^6)^{2/3}$, its broad contribution is bounded pointwise by $D^{-1} U^6 + C_Y D^{1/2} P^6$. Combining the two disjoint regions in (5.2) gives

$$\int |f_\tau|^6 \leq (1 + D^{-1}) \sum_{\sigma \text{ child}} \int |f_\sigma|^6 + C_Y D^{1/2} \int_{\mathcal{B}_\tau(T)} P_\tau^6.$$

The two regions together contribute only one copy of $D^{-1} \int U^6$.

To expand this recursion, set $q = 1 + D^{-1}$ and define

$$\mathcal{I}_k = \sum_{\tau \in \mathcal{P}_k} \int |f_\tau|^6, \quad \mathcal{G}_k = \sum_{\tau \in \mathcal{P}_k} \int_{\mathcal{B}_\tau(T)} P_\tau^6 \quad (0 \leq k < J).$$

Also define $\mathcal{I}_J$ by the same formula at the terminal level. Summing over level k gives $\mathcal{I}_k \leq q \mathcal{I}_{k+1} + C_Y D^{1/2} \mathcal{G}_k$. Successive substitution yields

$$\mathcal{I}_0 \leq q^J \mathcal{I}_J + C_Y D^{1/2} \sum_{k=0}^{J-1} q^k \mathcal{G}_k.$$

The coefficient of a level- k error is q^k , because it passes through precisely k ancestors, and the coefficient on each terminal integral is q^J . Since $\log(1 + D^{-1}) \leq D^{-1}$ and $J \leq D$,

$$1 \leq q^k \leq q^J \leq \exp(J/D) \leq \exp(1).$$

For a terminal interval, $S_\theta = \|f_\theta\|_\infty^2$, and hence

$$\mathcal{I}_J \leq \sum_{\theta \in \mathcal{P}_J} S_\theta^2 M_\theta \leq S_*^2 \sum_{\theta \in \mathcal{P}_J} M_\theta = S_*^2 M.$$

At any fixed level, $S_\tau \leq S_*$ and $\sum_\tau M_\tau = M$. Thus (5.3) and $T^2 = C_0^{1/2} D^{1/2}$ give

$$\begin{aligned} \mathcal{G}_k &\leq C_B D T^2 \sum_{\tau \in \mathcal{P}_k} S_\tau^2 M_\tau \\ &\leq C_B C_0^{1/2} D^{3/2} S_*^2 M. \end{aligned}$$

Inserting these bounds into the finite recursion gives

$$\int |f|^6 \leq \exp(1)(1 + C_Y C_B C_0^{1/2} J D^2) S_*^2 M \leq C D^3 S_*^2 M.$$

The last inequality uses $J \leq D$ and $D \geq 2$. This also proves the estimate when some nodes vanish; when $f = 0$, both sides are zero. $\square$

6 The endpoint estimate for a common terminal mass ratio

We now assume that the terminal pieces satisfy

$$\|f_\theta\|_2^2 = V \|f_\theta\|_\infty^2 \quad (\theta \in \mathcal{P}_J) \quad (6.1)$$

for some $V > 0$. Zero pieces are allowed. When the terminal depth is L , the same condition is imposed for $\theta \in \mathcal{P}_L$, and all masses S_*, S_τ, M_τ are computed from these level- L pieces. Then $M_\tau = V S_\tau$ at every node, by summing the condition over its terminal descendants. The proof in this section uses the strong broad estimate and the unabsorbed recursion (5.2); it does not use Proposition 5.3.

Fix $J \geq 1$. For $1 \leq L \leq J$, let $\mathfrak{C}_L$ consist of pairs (f, V) with $f \in \mathcal{S}(\mathbb{Q}_p^2) \setminus \{0\}$, $\text{supp } \widehat{f} \subseteq \Xi_L$, and (6.1) at terminal level L . In each such class, $S_*(f)$ is computed using its own terminal level. Define

$$X_J = \max \left\{ 1, \sup_{\substack{1 \leq L \leq J \\ (f, V) \in \mathfrak{C}_L}} \frac{\int |f|^6 d\mu}{V S_*(f)^3} \right\}. \quad (6.2)$$

The denominator is positive because $f \neq 0$ and $V > 0$. The supremum includes all smaller positive depths so that it can be applied after rescaling a node; terminal nodes, whose remaining depth is zero, will be estimated directly.

Lemma 6.1 (Finiteness and stability). *We have $1 \leq X_J \leq b^{2J} < \infty$. For every $(f, V) \in \mathfrak{C}_L$, each node satisfies*

$$\int |f_\tau|^6 d\mu \leq X_J V S_\tau^3. \quad (6.3)$$

Here the node masses are computed from the level- L terminal pieces. For every nonterminal node τ and every child σ_0 of τ , multiplying all terminal pieces contained in σ_0 by the same nonzero scalar preserves membership in $\mathfrak{C}_L$ and the value of V , provided the resulting function is nonzero.

Proof. There are b^L terminal intervals. The triangle inequality and Cauchy–Schwarz, followed by orthogonality for the squared L^2 norm, give

$$\begin{aligned}\|f\|_\infty &\leq \sum_{\theta \in \mathcal{P}_L} \|f_\theta\|_\infty \leq b^{L/2} S_*^{1/2}, \\ \|f\|_2^2 &= \sum_{\theta \in \mathcal{P}_L} \|f_\theta\|_2^2 = VS_*.\end{aligned}$$

It follows that

$$\int |f|^6 \leq \|f\|_\infty^4 \|f\|_2^2 \leq b^{2L} VS_*^3 \leq b^{2J} VS_*^3.$$

This proves finiteness before any use of the endpoint argument. In particular, it requires neither an extremizer for the supremum nor Proposition 5.3.

If $S_\tau = 0$, every terminal descendant has zero L^∞ norm, so $f_\tau = 0$ and (6.3) holds. Suppose next that $S_\tau > 0$ and $\tau \in \mathcal{P}_k$ with $k < L$. Rescale f_τ by (5.4). Its remaining depth is $L - k \in \{1, \dots, J\}$, and each terminal component $h_{\bar{\theta}}$ corresponds to one original f_θ inside τ . Formula (5.5) gives

$$\begin{aligned}\|h_{\bar{\theta}}\|_2^2 &= R_k^{-3/2} \|f_\theta\|_2^2 = R_k^{-3/2} V \|f_\theta\|_\infty^2 \\ &= V' \|h_{\bar{\theta}}\|_\infty^2, \quad V' = R_k^{-3/2} V > 0.\end{aligned}$$

The function h is nonzero, so $(h, V') \in \mathfrak{C}_{L-k}$, with $S_*(h) = S_\tau$. The definition of X_J therefore implies

$$R_k^{-3/2} \int |f_\tau|^6 = \int |h|^6 \leq X_J V' S_\tau^3 = X_J R_k^{-3/2} VS_\tau^3.$$

Cancelling the positive scale factor proves (6.3) at every nonterminal node. At a terminal node no rescaling to a depth-zero class is needed: directly,

$$\int |f_\theta|^6 \leq \|f_\theta\|_\infty^4 \|f_\theta\|_2^2 = VS_\theta^3 \leq X_J VS_\theta^3.$$

Finally, let $c \neq 0$ be a scalar multiplying the descendants of a chosen child. Their Fourier supports remain in the same disjoint terminal intervals, so the terminal projections of the new function are exactly cf_θ for the scaled descendants and f_θ for all others. For every scaled piece,

$$\|cf_\theta\|_2^2 = |c|^2 \|f_\theta\|_2^2 = V|c|^2 \|f_\theta\|_\infty^2 = V\|cf_\theta\|_\infty^2.$$

Zero pieces continue to satisfy this equality, and scalar multiplication preserves the Schwartz–Bruhat class and the required support. Thus the nonzero resulting function remains in $\mathfrak{C}_L$ with the same V . $\square$

Lemma 6.2 (An estimate for the mixed term). *Let $D \geq \max\{2, 4J \log p\}$ and suppose $X_J \geq D$. Put*

$$T_X = (X_J/D)^{1/8}, \quad C_M = C_B^{2/3} + b^{-4}.$$

For every $(f, V) \in \mathfrak{C}_L$, $1 \leq L \leq J$, and every nonterminal node,

$$\int Q_\tau d\mu \leq C_M \sqrt{DX_J} VS_\tau^3. \quad (6.4)$$

Proof. The assertion holds directly if $S_\tau = 0$. Otherwise write $A = A_\tau$, $P = P_\tau$ and $U = U_\tau$. Since U^6 is the sum of the child sixth powers and $A \leq U$, (6.3) gives

$$\begin{aligned}\int A^6 &\leq \int U^6 = \sum_{\sigma \text{ child}} \int |f_\sigma|^6 \\ &\leq X_J V \sum_{\sigma \text{ child}} S_\sigma^3 \leq X_J VS_\tau^3.\end{aligned}$$

The last inequality follows by expanding the cube of $S_\tau = \sum_\sigma S_\sigma$: the remaining terms are all nonnegative. This also covers children that are terminal or have zero mass, by Lemma 6.1.

Initially let $T \geq 1$ be arbitrary. On $\mathcal{B}_\tau(T)$, write $Q_\tau = A^2 P^4 = (A^6)^{1/3} (P^6)^{2/3}$. Hölder's inequality with conjugate exponents 3 and 3/2 gives

$$\int_{\mathcal{B}_\tau(T)} Q_\tau \leq \left(\int_{\mathcal{B}_\tau(T)} A^6 \right)^{1/3} \left(\int_{\mathcal{B}_\tau(T)} P^6 \right)^{2/3}.$$

Here (5.3) applies at depth $L \leq J$, with $M_\tau = VS_\tau$. Substituting it and the preceding sixth-moment bound yields

$$\begin{aligned} \int_{\mathcal{B}_\tau(T)} Q_\tau &\leq (X_J VS_\tau^3)^{1/3} (C_B DT^2 VS_\tau^3)^{2/3} \\ &= C_B^{2/3} X_J^{1/3} D^{2/3} T^{4/3} VS_\tau^3. \end{aligned}$$

In the final line, the powers of VS_τ^3 add to one. On the complement, $P < U/(Tb)$ and $A \leq U$, so

$$Q_\tau = A^2 P^4 \leq (Tb)^{-4} U^6, \quad \int_{\mathcal{B}_\tau(T)^c} Q_\tau \leq b^{-4} X_J T^{-4} VS_\tau^3.$$

To balance the two parameter factors, solve $X_J^{1/3} D^{2/3} T^{4/3} = X_J T^{-4}$. This is equivalent to $T^{16/3} = (X_J/D)^{2/3}$, or $T^8 = X_J/D$. Thus we take the stated $T = T_X$. It is finite by Lemma 6.1 and at least one because $X_J \geq D$, so the broad estimate permits this choice. More explicitly,

$$\begin{aligned} X_J^{1/3} D^{2/3} T_X^{4/3} &= X_J^{1/3} D^{2/3} (X_J/D)^{1/6} = \sqrt{DX_J}, \\ X_J T_X^{-4} &= X_J (X_J/D)^{-1/2} = \sqrt{DX_J}. \end{aligned}$$

Adding the two contributions gives (6.4) with $C_M = C_B^{2/3} + b^{-4}$. The same T_X works for every function and node included in the supremum defining X_J . $\square$

Lemma 6.3 (Rebalancing child masses). *Under the assumptions of Lemma 6.2, let*

$$\Delta_\tau = S_\tau^3 - \sum_{\sigma \text{ child of } \tau} S_\sigma^3.$$

Then

$$\int Q_\tau d\mu \leq \frac{8}{3} C_M \sqrt{DX_J} V \Delta_\tau. \quad (6.5)$$

Proof. All child masses are nonnegative and add to S_τ , so $\Delta_\tau \geq 0$. Set $m = \max_\sigma S_\sigma$ and $s = S_\tau - m$. If $s = 0$, at most one child is nonzero. Every product defining Q_τ then has a zero factor, and $\Delta_\tau = m^3 - m^3 = 0$. This includes the zero-mass node $m = s = 0$.

Suppose first that $s \geq m > 0$. Then $m \leq S_\tau/2$, and each child mass satisfies $S_\sigma^3 \leq m^2 S_\sigma$. Summing gives

$$\sum_\sigma S_\sigma^3 \leq m^2 \sum_\sigma S_\sigma = m^2 S_\tau \leq S_\tau^3/4.$$

Thus $\Delta_\tau \geq (3/4)S_\tau^3$. Lemma 6.2 consequently gives the bound with coefficient 4/3, which is at most the asserted 8/3.

It remains to treat $0 < s < m$. Choose the child σ_0 of mass m , put $c = \sqrt{s/m} \in (0, 1)$, and form $f' = f + (c - 1)f_{\sigma_0}$. This multiplies exactly the terminal descendants of σ_0 by c . By Lemma 6.1, the new function remains in $\mathfrak{C}_L$ with the same V ; it is nonzero since its mass inside τ is positive. The new child masses at this node satisfy

$$S'_{\sigma_0} = c^2 m = s, \quad S'_\sigma = S_\sigma \ (\sigma \neq \sigma_0), \quad S'_\tau = c^2 m + s = 2s.$$

We compare every ordered product in the definition of Q_τ , without identifying the child of largest pointwise amplitude. For distinct σ, σ' , its exact multiplier is

$$|f'_\sigma|^4 |f'_{\sigma'}|^2 = \begin{cases} c^4 |f_\sigma|^4 |f_{\sigma'}|^2, & \sigma = \sigma_0, \\ c^2 |f_\sigma|^4 |f_{\sigma'}|^2, & \sigma' = \sigma_0, \\ |f_\sigma|^4 |f_{\sigma'}|^2, & \sigma, \sigma' \neq \sigma_0. \end{cases}$$

The pair cannot contain σ_0 in both positions because its indices are distinct. Since $0 < c < 1$, each multiplier is at least c^4 . Taking the maximum over all ordered pairs gives $Q_\tau(f') \geq c^4 Q_\tau(f)$, even if the pair attaining the maximum changes after scaling. Applying (6.4) to f' gives

$$\begin{aligned} \int Q_\tau(f) &\leq c^{-4} \int Q_\tau(f') \leq c^{-4} C_M \sqrt{DX_J} V(2s)^3 \\ &= 8C_M \sqrt{DX_J} Vm^2 s, \end{aligned}$$

where $c^{-4} = (m/s)^2$. On the other hand, the remaining child masses add to s , so $\sum_{\sigma \neq \sigma_0} S_\sigma^3 \leq s^3$. Consequently,

$$\begin{aligned} \Delta_\tau &= (m+s)^3 - m^3 - \sum_{\sigma \neq \sigma_0} S_\sigma^3 \\ &\geq (m+s)^3 - m^3 - s^3 = 3m^2 s + 3ms^2 \geq 3m^2 s. \end{aligned}$$

Combining the last two bounds proves (6.5). No lower bound on s/m was used: c remains nonzero for every $s > 0$, and the case $s = 0$ was already treated directly. $\square$

Theorem 6.4 (Endpoint mixed estimate). *Let $J \geq 1$, $D \geq \max\{2, 4J \log p\}$, and $1 \leq L \leq J$. For every $(f, V) \in \mathfrak{C}_L$,*

$$\int_{\mathbb{Q}_p^2} |f|^6 d\mu \leq C_E DVS_*(f)^3, \quad C_E = 1 + 4K_0^2, \quad K_0 = \frac{8}{3} C_0 C_M. \quad (6.6)$$

In particular, C_E depends only on p , and is independent of J, L, D, f and V .

Proof. If $X_J < D$, its definition gives $\int |f|^6 \leq DVS_*^3 \leq C_E DVS_*^3$ for every required pair. Suppose henceforth that $X_J \geq D$, so Lemmas 6.2 and 6.3 apply.

Sum the coefficient-one recursion (5.2) over level k . Every level- $(k+1)$ node occurs exactly once as a child, giving

$$\sum_{\tau \in \mathcal{P}_k} \int |f_\tau|^6 \leq \sum_{\sigma \in \mathcal{P}_{k+1}} \int |f_\sigma|^6 + C_0 \sum_{\tau \in \mathcal{P}_k} \int Q_\tau.$$

Substitute this inequality successively for $k = 0, \dots, L-1$. All intermediate sixth moments cancel, leaving the finite-tree inequality

$$\int |f|^6 \leq \sum_{\theta \in \mathcal{P}_L} \int |f_\theta|^6 + C_0 \sum_{k=0}^{L-1} \sum_{\tau \in \mathcal{P}_k} \int Q_\tau.$$

At the terminal level, $\int |f_\theta|^6 \leq VS_\theta^3$. For the mixed terms, the cubic mass differences telescope with the same parent-child indexing:

$$\sum_{\tau \in \mathcal{P}_k} \Delta_\tau = \sum_{\tau \in \mathcal{P}_k} S_\tau^3 - \sum_{\sigma \in \mathcal{P}_{k+1}} S_\sigma^3.$$

Summing over all nonterminal levels gives

$$\sum_{k=0}^{L-1} \sum_{\tau \in \mathcal{P}_k} \Delta_\tau = S_*^3 - \sum_{\theta \in \mathcal{P}_L} S_\theta^3 \leq S_*^3.$$

Every intermediate mass occurs once positively and once negatively. These are finite sums, and zero-mass nodes contribute zero. Lemma 6.3 therefore implies

$$\int |f|^6 \leq V \sum_{\theta \in \mathcal{P}_L} S_\theta^3 + K_0 \sqrt{DX_J} V \left(S_*^3 - \sum_{\theta \in \mathcal{P}_L} S_\theta^3 \right).$$

Both cubic quantities on the right are bounded by S_*^3 , and $VS_*^3 > 0$. Dividing and using these bounds yields

$$\frac{\int |f|^6}{VS_*^3} \leq 1 + K_0 \sqrt{DX_J}.$$

The quantities X_J and D were fixed before choosing L or f . Hence the inequality has the same right side for every pair in (6.2), and taking their supremum preserves the bound. The right side is also at least one, so the additional maximum with one gives

$$X_J \leq 1 + K_0 \sqrt{DX_J}. \quad (6.7)$$

This is an inequality for a finite positive number, by Lemma 6.1. Since $X_J \geq D \geq 2$, we have $X_J - 1 \geq X_J/2$. Subtracting one in (6.7), dividing by $\sqrt{X_J} > 0$, and squaring give

$$\frac{X_J}{2} \leq K_0 \sqrt{DX_J}, \quad \frac{\sqrt{X_J}}{2} \leq K_0 \sqrt{D}, \quad X_J \leq 4K_0^2 D \leq C_E D.$$

Together with the case $X_J < D$, this proves the theorem for every $1 \leq L \leq J$. All applications of the supremum involved finite constants and actual class members; no maximizing function or limiting argument is required. $\square$

Remark 6.5. The hypothesis (6.1) is used in the rescaling and rebalancing argument and is retained in Theorem 6.4. For unrestricted Schwartz–Bruhat functions, Proposition 5.3 gives the weaker factor D^3 . The exact torus embedding below satisfies (6.1) for all complex coefficient sequences, so no coefficient restriction remains in Theorem 1.1.

7 Transfer to the torus

The transfer uses exact character orthogonality. The radius of the spatial ball is chosen so that the resulting congruences between integer frequencies are equivalent to equalities.

Lemma 7.1. *Let $J \geq 1$, $N_0 = b^J = p^{2J}$, $R = R_J = N_0^2$, and $B = B(0, R^2) \subseteq \mathbb{Q}_p^2$. Given $a \in \mathbb{C}^{N_0}$, set*

$$F(x, t) = \mathbf{1}_B(x, t) \sum_{n=1}^{N_0} a_n \chi(nx + n^2 t).$$

Then $F \in \mathcal{S}(\mathbb{Q}_p^2)$, $\text{supp } \widehat{F} \subseteq \Xi_J$, and

$$\sum_{\theta \in \mathcal{P}_J} \|F_\theta\|_\infty^2 = \sum_{n=1}^{N_0} |a_n|^2, \quad \|F\|_2^2 = \mu(B) \sum_{n=1}^{N_0} |a_n|^2. \quad (7.1)$$

Moreover,

$$\int_{\mathbb{Q}_p^2} |F|^6 d\mu = \mu(B) \int_{\mathbb{T}^2} |E_{N_0} a(x, t)|^6 dx dt. \quad (7.2)$$

Each nonzero terminal component satisfies

$$\|F_\theta\|_2^2 = \mu(B) \|F_\theta\|_\infty^2.$$

Proof. The character factors and the indicator of B are locally constant, and B is compact. Hence $F \in \mathcal{S}(\mathbb{Q}_p^2)$. The annihilator of B is $B^\perp = B(0, R^{-2})$. By (2.1) and the modulation identity,

$$\begin{aligned} \mathbf{1}_B \widehat{\chi(nx + n^2 t)}(\xi, \eta) &= \widehat{\mathbf{1}_B}(\xi - n, \eta - n^2) \\ &= \mu(B) \mathbf{1}_{B(0, R^{-2})}(\xi - n, \eta - n^2). \end{aligned}$$

Consequently,

$$\widehat{F}(\xi, \eta) = \mu(B) \sum_{n=1}^{N_0} a_n \mathbf{1}_{(n, n^2) + B(0, R^{-2})}(\xi, \eta).$$

For a point $(\xi, \eta) = (n + u, n^2 + v)$ in the n th ball, $|u|_p, |v|_p \leq R^{-2}$ and $n \in \mathbb{Z}_p$. Thus $\xi \in \mathbb{Z}_p$, and the ultrametric inequality gives

$$\begin{aligned} |\eta - \xi^2|_p &= |v - 2nu - u^2|_p \\ &\leq \max\{|v|_p, |2n|_p |u|_p, |u|_p^2\} \leq R^{-2} \leq R^{-1}. \end{aligned}$$

This proves $\text{supp } \widehat{F} \subseteq \Xi_J$.

To identify the terminal projections, let $\theta_n = n + p^{2J} \mathbb{Z}_p$. The integers $1, \dots, N_0$ form a complete set of residues modulo $p^{2J} = N_0$, so these are precisely the distinct members of $\mathcal{P}_J$. Since $R^{-2} \leq R^{-1/2}$, the first-coordinate projection of the n th Fourier ball lies entirely in θ_n . Multiplication of $\widehat{F}$ by $\mathbf{1}_{\theta_n}(\xi)$ therefore retains only its n th summand. Fourier inversion yields

$$F_{\theta_n}(x, t) = a_n \mathbf{1}_B(x, t) \chi(nx + n^2 t).$$

Because $|\chi| = 1$ and $\mu(B) = (R^2)^2 = R^4$, we have

$$\|F_{\theta_n}\|_\infty^2 = |a_n|^2, \quad \|F_{\theta_n}\|_2^2 = \int_B |a_n|^2 d\mu = R^4 |a_n|^2.$$

These formulas include $a_n = 0$. Summing the first identity gives the first assertion in (7.1); orthogonality of the terminal projections gives the second:

$$\|F\|_2^2 = \sum_{n=1}^{N_0} \|F_{\theta_n}\|_2^2 = \mu(B) \sum_{n=1}^{N_0} |a_n|^2.$$

The same identities also give the common terminal mass ratio $V = \mu(B)$.

We now compare the sixth moments. For a six-tuple $\mathbf{n} = (n_1, \dots, n_6)$, with $1 \leq n_i \leq N_0$, put

$$\begin{aligned} u(\mathbf{n}) &= n_1 + n_2 + n_3 - n_4 - n_5 - n_6, \\ v(\mathbf{n}) &= n_1^2 + n_2^2 + n_3^2 - n_4^2 - n_5^2 - n_6^2, \\ c(\mathbf{n}) &= a_{n_1} a_{n_2} a_{n_3} \overline{a_{n_4} a_{n_5} a_{n_6}}. \end{aligned}$$

Expanding $F^3 \overline{F}^3$ and using $\mathbf{1}_B^6 = \mathbf{1}_B$, we obtain

$$\int_{\mathbb{Q}_p^2} |F|^6 d\mu = \sum_{\mathbf{n}} c(\mathbf{n}) \int_B \chi(u(\mathbf{n})x + v(\mathbf{n})t) d\mu(x, t).$$

There are N_0^6 terms, each integrated over a set of finite measure, so exchanging this sum and the integral is legitimate. By character orthogonality,

$$\frac{1}{\mu(B)} \int_B \chi(ux + vt) d\mu(x, t) = \mathbf{1}_{B^\perp}(u, v) = \mathbf{1}_{\{|u|_p \leq R^{-2}, |v|_p \leq R^{-2}\}}.$$

Here u, v are integers and $R^{-2} = p^{-8J}$. For an integer w , the condition $|w|_p \leq p^{-8J}$ is equivalent to $p^{8J} \mid w$, including $w = 0$. On the other hand, the ordinary absolute values of the two defects satisfy

$$|u| \leq 3N_0 < R^2, \quad |v| \leq 3N_0^2 = 3R < R^2,$$

because $N_0 \geq 9$ and $R \geq 81$. The only multiple of $R^2 = p^{8J}$ with absolute value smaller than R^2 is zero. Thus

$$\frac{1}{\mu(B)} \int_B \chi(ux + vt) d\mu(x, t) = \mathbf{1}_{\{u=0, v=0\}}.$$

The torus integral gives exactly the same indicator, since

$$\int_{\mathbb{T}^2} e(ux + vt) dx dt = \left(\int_0^1 e(ux) dx \right) \left(\int_0^1 e(vt) dt \right) = \mathbf{1}_{\{u=0, v=0\}}.$$

The finite expansion of $|E_{N_0}a|^6$ has the same coefficients $c(\mathbf{n})$. Hence

$$\frac{1}{\mu(B)} \int_{\mathbb{Q}_p^2} |F|^6 d\mu = \sum_{\substack{\mathbf{n} \\ u(\mathbf{n})=v(\mathbf{n})=0}} c(\mathbf{n}) = \int_{\mathbb{T}^2} |E_{N_0}a|^6 dx dt,$$

which proves (7.2). The argument compares the coefficients term by term and therefore imposes no sign or phase restriction on a . $\square$

Proof of Theorem 1.1. First suppose that $N_0 = b^J = p^{2J}$ with $J \geq 1$ and construct F as in Lemma 7.1. The all-zero coefficient sequence gives both sides of (1.2) equal to zero. Otherwise $\|F\|_2^2 > 0$, so $F \neq 0$, and the lemma shows that $(F, \mu(B)) \in \mathfrak{C}_J$. In this class,

$$V = \mu(B), \quad S_*(F) = \sum_{n=1}^{N_0} |a_n|^2.$$

Set $D = \log(2R) = \log 2 + 4J \log p$. Then $D \geq 4J \log p$, and $J \geq 1$ implies $D \geq \log 162 > 2$, so all hypotheses of Theorem 6.4 hold. Writing $C_E(p)$ for its constant, which depends only on the chosen prime, its application to F gives

$$\int_{\mathbb{Q}_p^2} |F|^6 d\mu \leq C_E(p) \log(2R) \mu(B) \left(\sum_{n=1}^{N_0} |a_n|^2 \right)^3.$$

Replace the left side by (7.2), divide by the strictly positive quantity $\mu(B)$, and use $R = N_0^2$. We obtain

$$\int_{\mathbb{T}^2} |E_{N_0}a|^6 \leq C_E(p) \log(2N_0^2) \left(\sum_{n=1}^{N_0} |a_n|^2 \right)^3. \quad (7.3)$$

Now let $N \geq 2$ be arbitrary. Choose the least integer $J \geq 1$ for which $N \leq b^J$ and write $N_0 = b^J = p^{2J}$. Minimality gives $b^{J-1} < N$, and hence $N_0 < bN = p^2N$. Define $\tilde{a} \in \mathbb{C}^{N_0}$ by

$$\tilde{a}_n = \begin{cases} a_n, & 1 \leq n \leq N, \\ 0, & N < n \leq N_0. \end{cases}$$

Directly from the definitions,

$$E_{N_0}\tilde{a} = E_N a, \quad \sum_{n=1}^{N_0} |\tilde{a}_n|^2 = \sum_{n=1}^N |a_n|^2.$$

Put $\Lambda_p = 2 + 4 \log p / \log 2$. Since $\log(2N) \geq \log 2$, we have

$$\begin{aligned} \log(2N_0^2) &\leq \log 2 + 4 \log p + 2 \log N \\ &\leq 2 \log(2N) + 4 \log p \leq \Lambda_p \log(2N). \end{aligned}$$

Apply (7.3) to $\tilde{a}$ and substitute these identities. Taking sixth roots yields

$$\|E_N a\|_6 \leq (\Lambda_p C_E(p))^{1/6} [\log(2N)]^{1/6} \|a\|_{\ell^2}.$$

If $N = 1$, then $|E_1 a(x, t)| = |a_1|$ everywhere on the probability torus. Thus $\|E_1 a\|_6 = |a_1|$, and (1.2) holds for every $N \geq 1$ with a constant depending only on p , namely

$$C_p = \max\{(\Lambda_p C_E(p))^{1/6}, (\log 2)^{-1/6}\}.$$

This proves the torus upper bound using every fixed odd prime $p > 2$. To obtain the absolute constant in Theorem 1.1, fix $p = 3$ once and for all. Then

$$C = C_3 = \max\{[\Lambda_3 C_E(3)]^{1/6}, (\log 2)^{-1/6}\}, \quad \Lambda_3 = 2 + \frac{4 \log 3}{\log 2},$$

is independent of N and the coefficients. For $N \geq 2$, we also have $\log(2N) \leq 2 \log N$. Taking the supremum over nonzero coefficient sequences gives $K_{\mathbb{T}}(N) \lesssim (\log N)^{1/6}$. The reverse inequality is the classical lower bound recalled below, which completes the asserted two-sided comparison. $\square$

Corollary 7.2 (Weighted sixth-order resonances). *For every $N \geq 1$ and $a \in \mathbb{C}^N$,*

$$\begin{aligned} &\sum_{\substack{1 \leq n_1, \dots, n_6 \leq N \\ n_1 + n_2 + n_3 = n_4 + n_5 + n_6 \\ n_1^2 + n_2^2 + n_3^2 = n_4^2 + n_5^2 + n_6^2}} a_{n_1} a_{n_2} a_{n_3} \overline{a_{n_4} a_{n_5} a_{n_6}} \\ &= \|E_N a\|_{L^6(\mathbb{T}^2)}^6 \leq C^6 \log(2N) \left(\sum_{n=1}^N |a_n|^2 \right)^3. \end{aligned} \tag{7.4}$$

In particular, the complete sum is real and nonnegative, even when its individual terms are not.

Proof. With $u(\mathbf{n})$, $v(\mathbf{n})$ and $c(\mathbf{n})$ defined as in the transfer proof, now using indices in $\{1, \dots, N\}$, the finite expansion is

$$|E_N a(x, t)|^6 = \sum_{\mathbf{n} \in \{1, \dots, N\}^6} c(\mathbf{n}) e(u(\mathbf{n})x + v(\mathbf{n})t).$$

Integrating each term over $\mathbb{T}^2$ removes every six-tuple with a nonzero defect and gives the equality in (7.4). Raising (1.2) to the sixth power gives its inequality, since $\|a\|_{\ell^2}^6 = (\sum_n |a_n|^2)^3$. The equality with the integral of the nonnegative function $|E_N a|^6$ proves the final assertion. $\square$

Bourgain's major-arc calculation [1, p. 118, (2.50)–(2.51)], with his upper index taken to be $N - 1$, gives a sixth-moment lower bound of order $N^3 \log N$ for the sum indexed by $0, \dots, N - 1$. The identity

$$\sum_{n=1}^N e(nx + n^2 t) = e(x + t) \sum_{m=0}^{N-1} e(m(x + 2t) + m^2 t)$$

transfers this bound to our indexing: the factor has modulus one and the torus map $(x, t) \mapsto (x + 2t, t)$ preserves measure. For the all-one coefficient vector $\mathbf{1}_N$, this gives

$$K_{\mathbb{T}}(N)^6 \geq \frac{\|E_N \mathbf{1}_N\|_6^6}{\|\mathbf{1}_N\|_{\ell^2}^6} \geq \frac{c N^3 \log N}{N^3} = c \log N$$

for all sufficiently large N . The remaining finite range $N \geq 2$ is covered by decreasing the constant, since $K_{\mathbb{T}}(N) \geq 1$ by testing a single nonzero coefficient. Consequently,

$$K_{\mathbb{T}}(N) \asymp (\log N)^{1/6} \quad (N \geq 2).$$

This comparison establishes the sharp logarithmic exponent. It does not assert that the normalized restriction constant converges or identify a leading asymptotic constant.

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